



Building RAG Agents with LLMs

Jae Y. CHOI

NVIDIA DLI Certified Instructor & University Ambassador

본 자료의 저작권은 Jae Y. CHOI (jaeyeong2022@gmail.com)에게 있습니다.

본 자료는 자유롭게 이용할 수 있으나, 사용 시 저작자 및 출처를 반드시 표기해 주시기 바랍니다.

Copyright © Jae Y. CHOI (jaeyeong2022@gmail.com).

This material may be freely used, provided that the author and source are properly credited.

Course Objectives

Fundamentals of LLM & Agentic AI

- Introduction to LLM
- LLM Reasoning and Agentic AI
- LLM Training
- Evolution of AI Models
- GPU Infrastructure





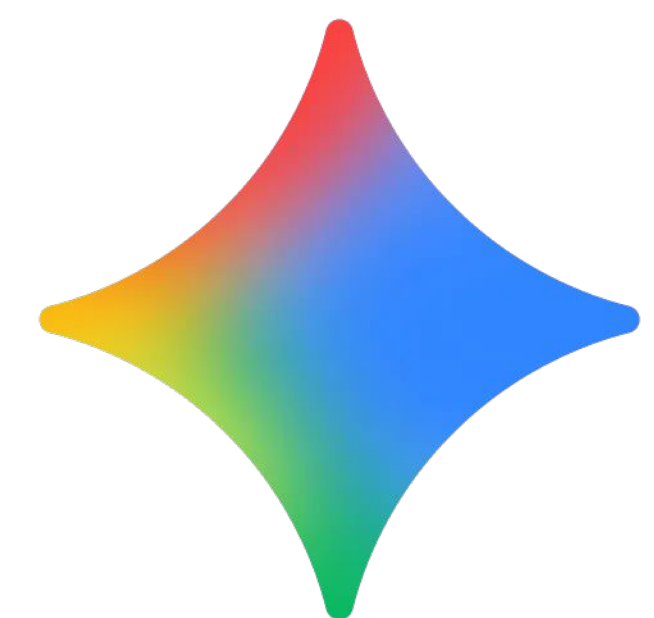
Building RAG Agents with LLMs

Part 1. Introduction to LLM

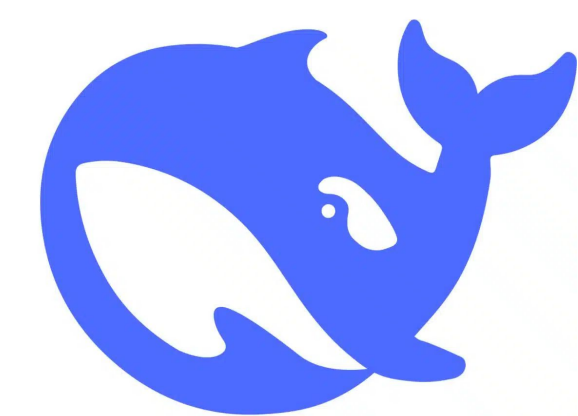
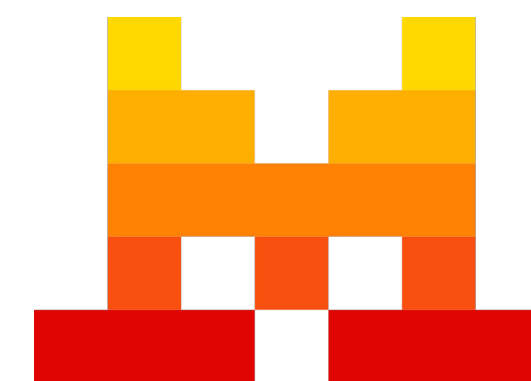
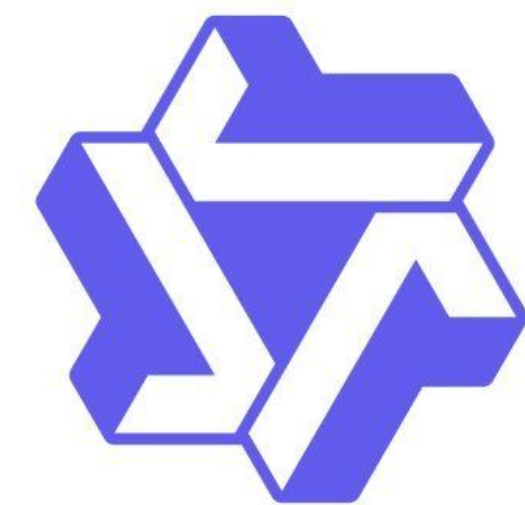
Introduction to LLM

What is the Large Language Model?

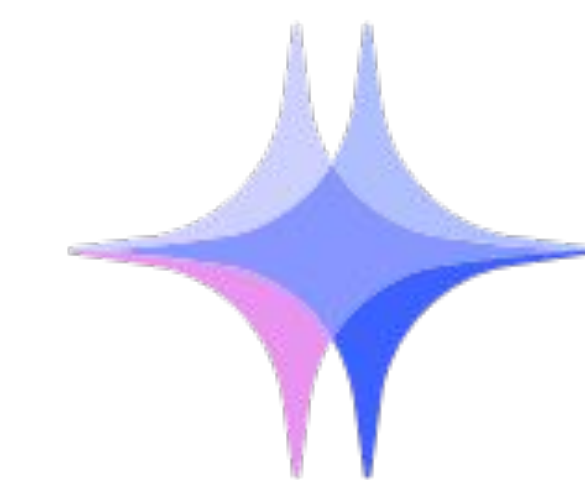
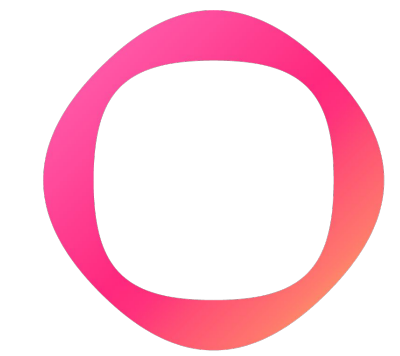
Proprietary(Closed) Models



Global Opensource Models



Korea AI Models

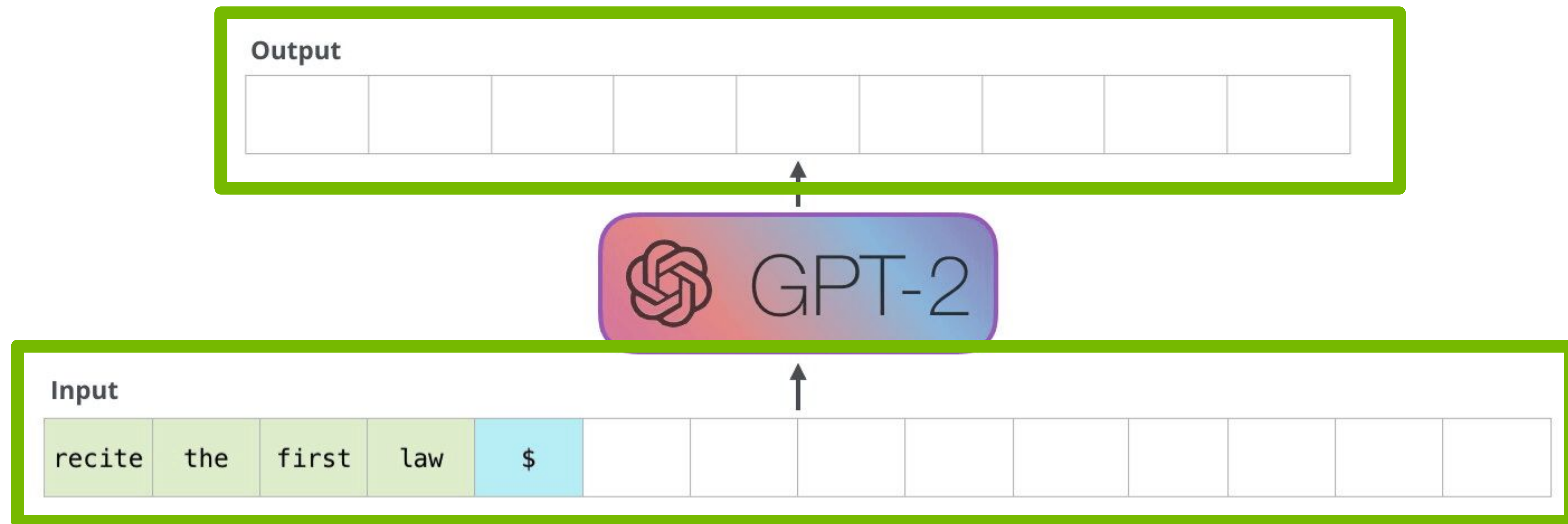


Too Many Models!!

Introduction to LLM

What is the Large Language Model?

User Area

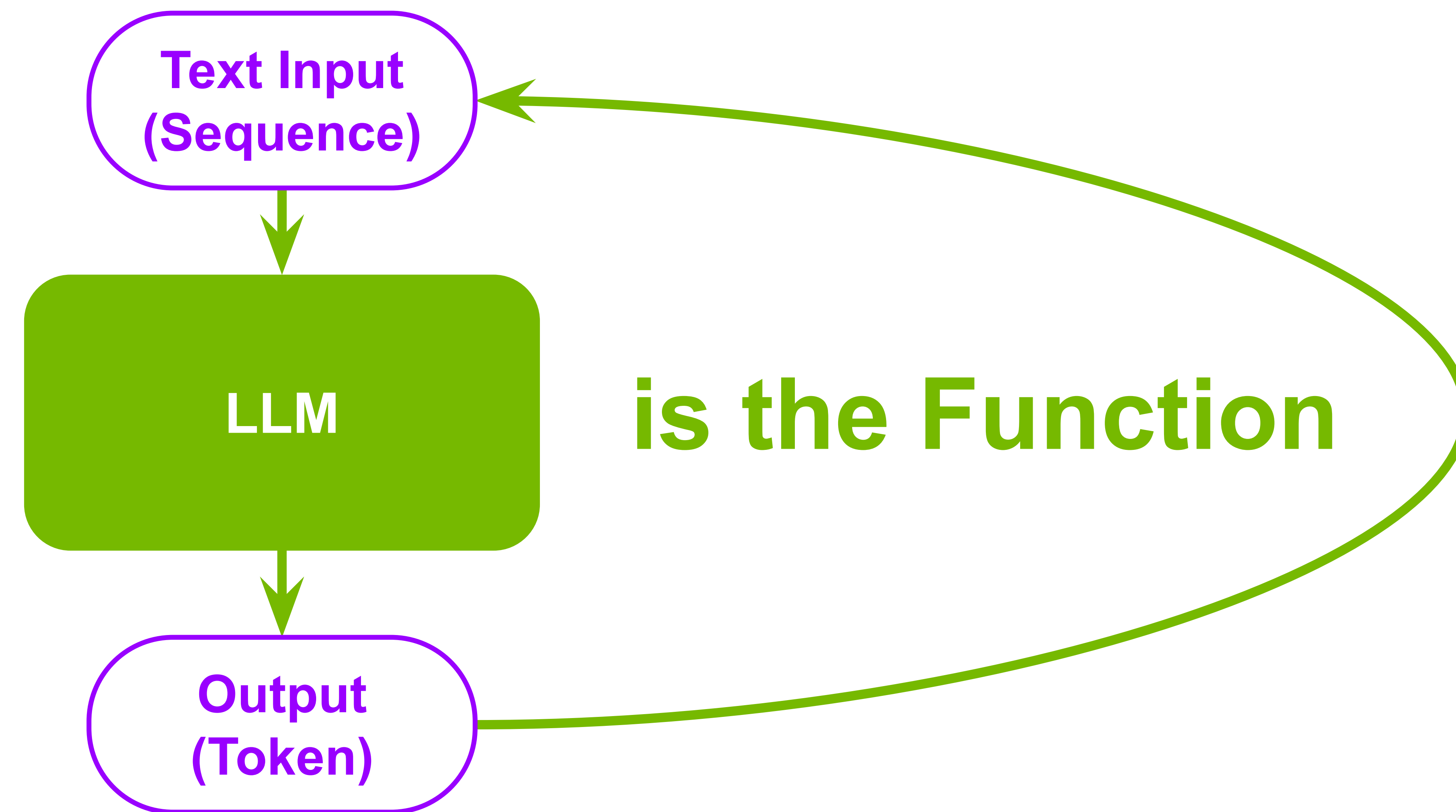


What's under the hood?
Auto Regressive (AR)

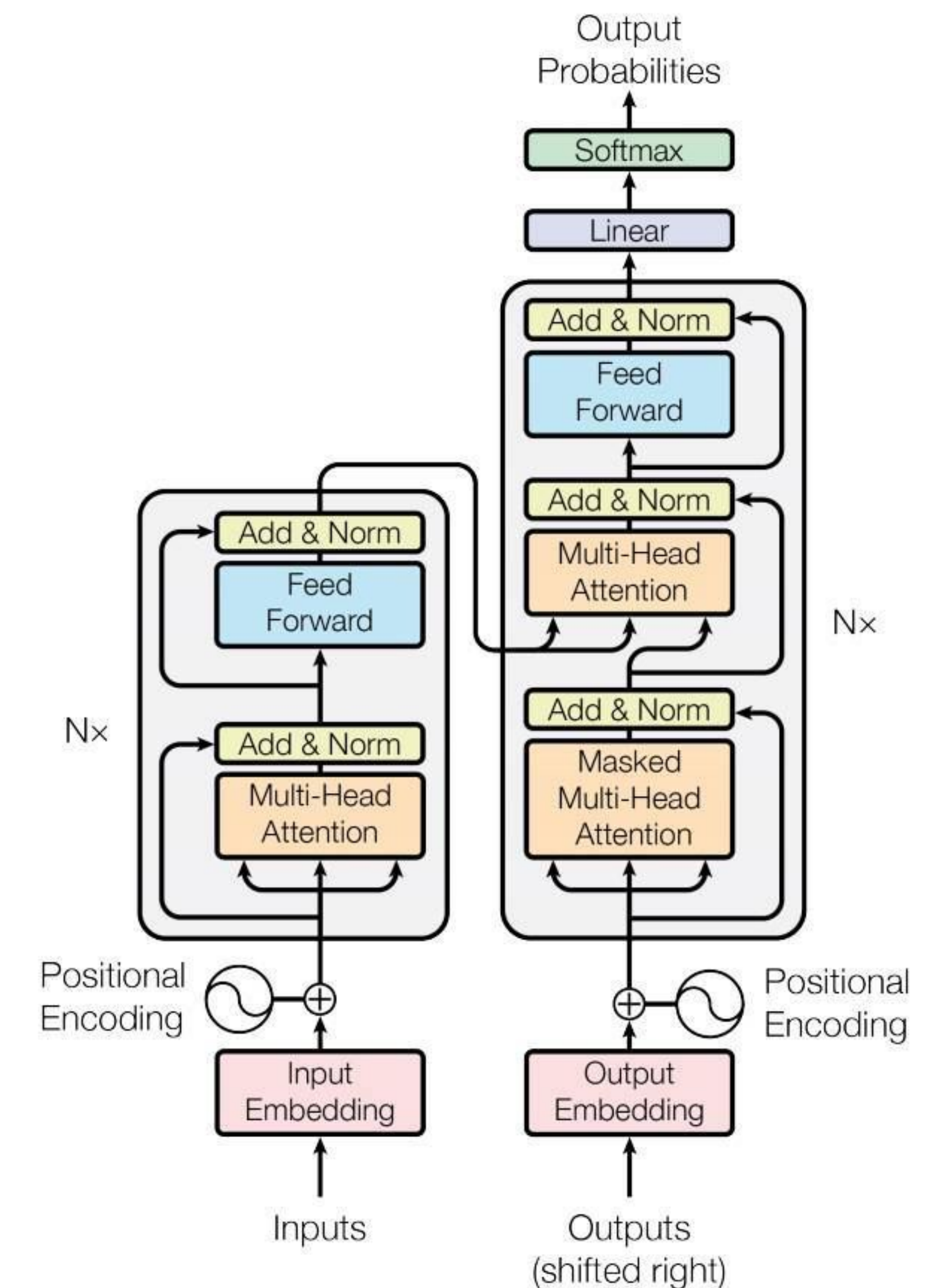
Introduction to LLM

What is the Large Language Model?

New insight of view



Baseline Architecture
→ Transformer

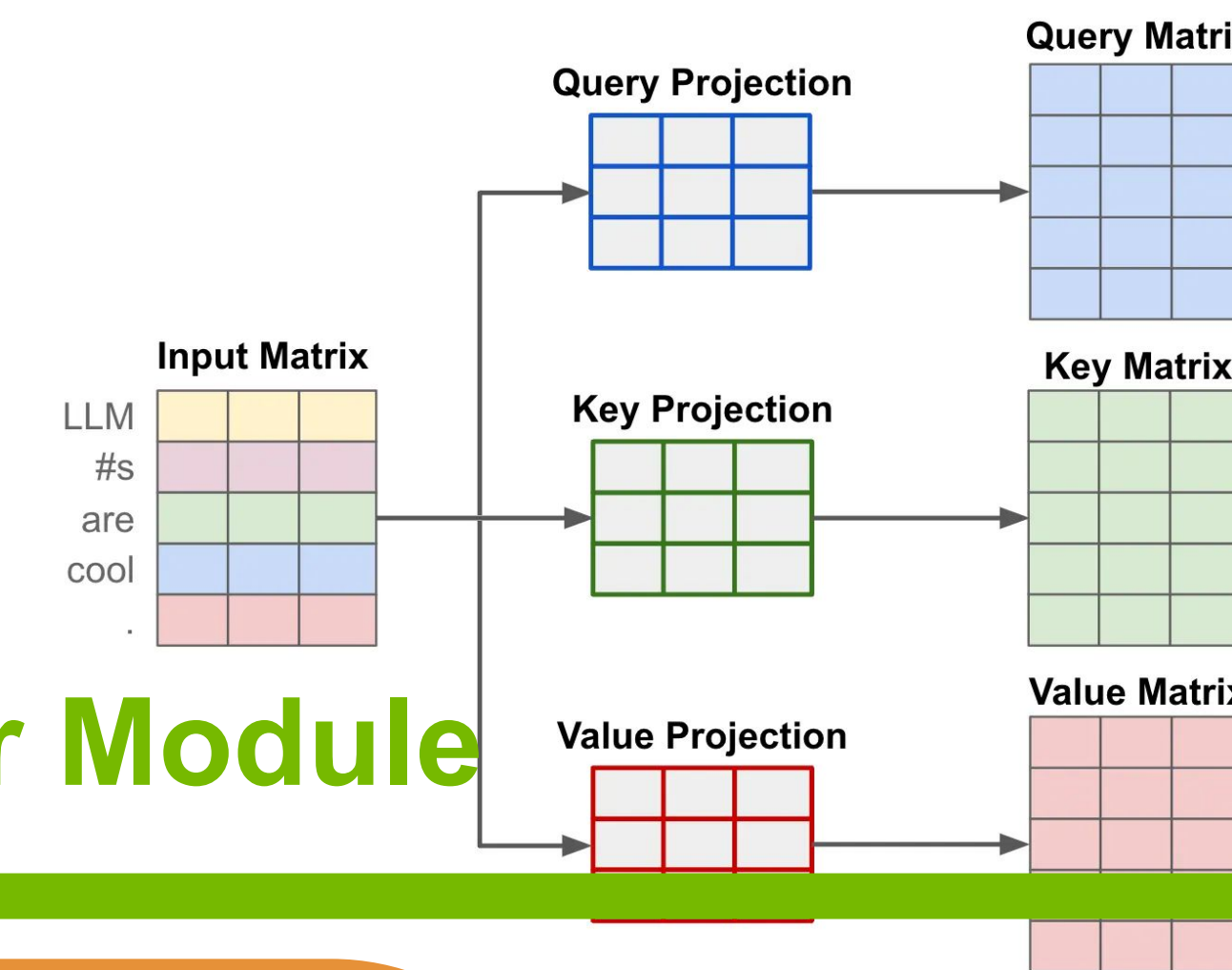
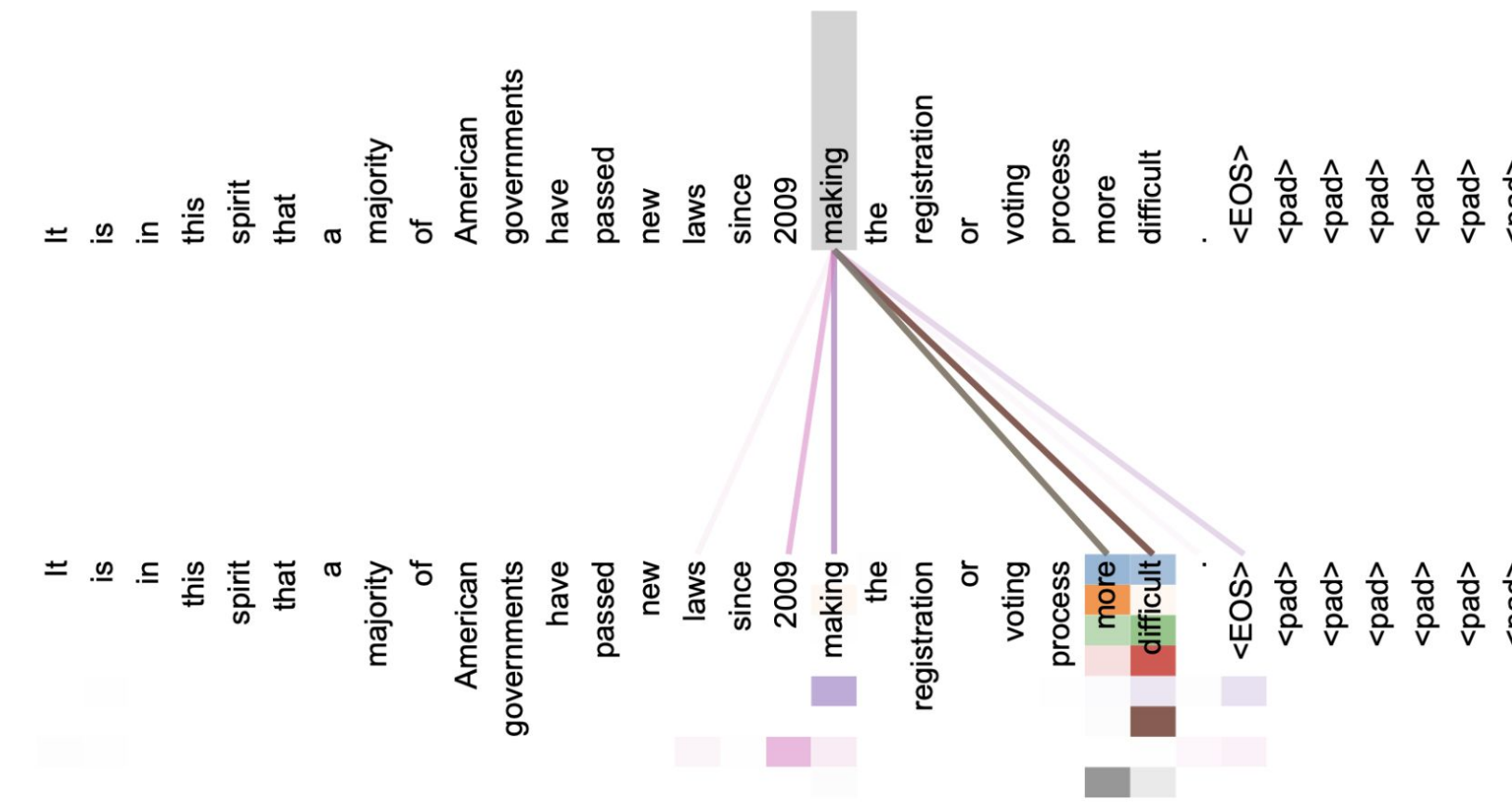
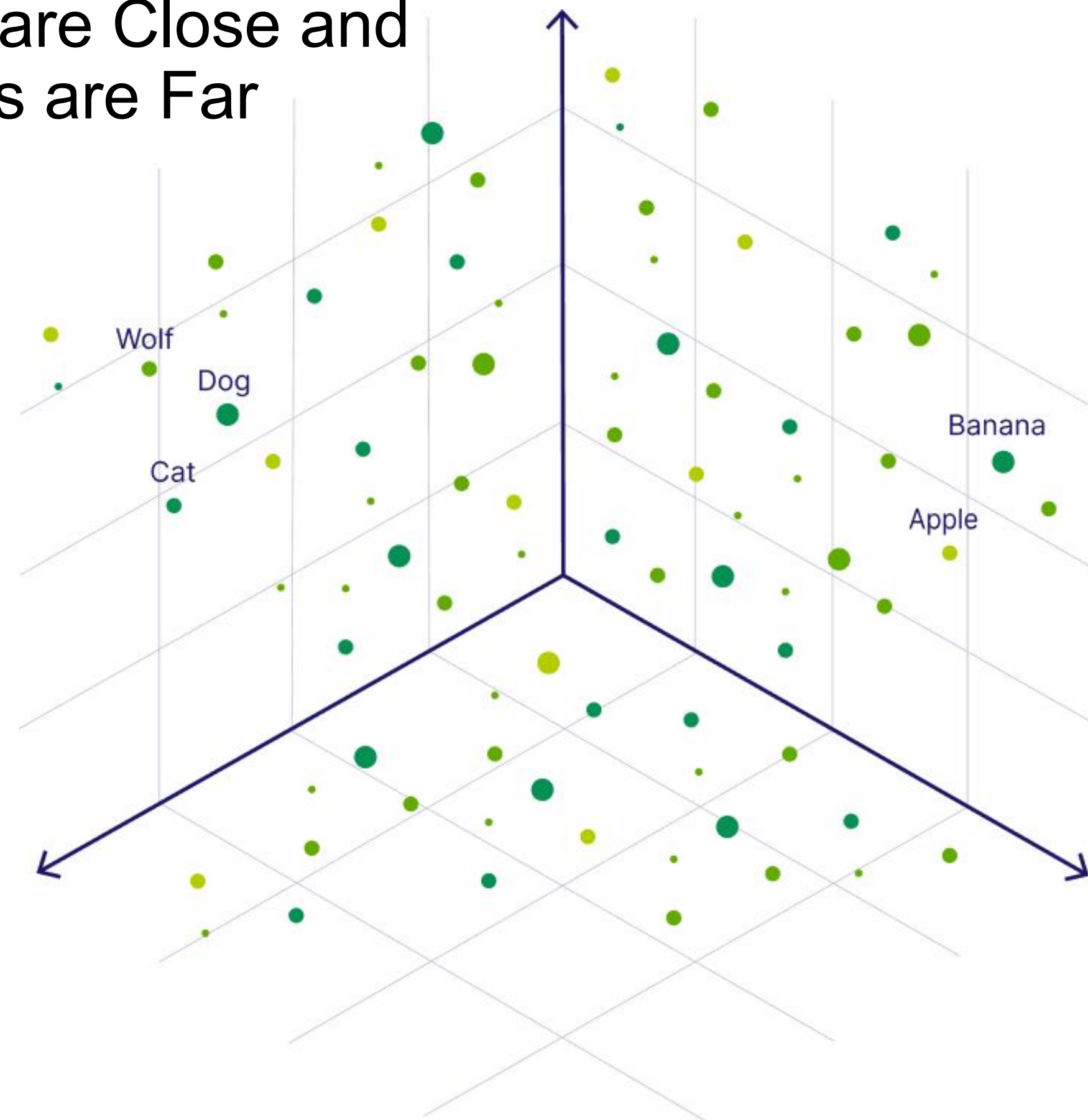


Introduction to LLM

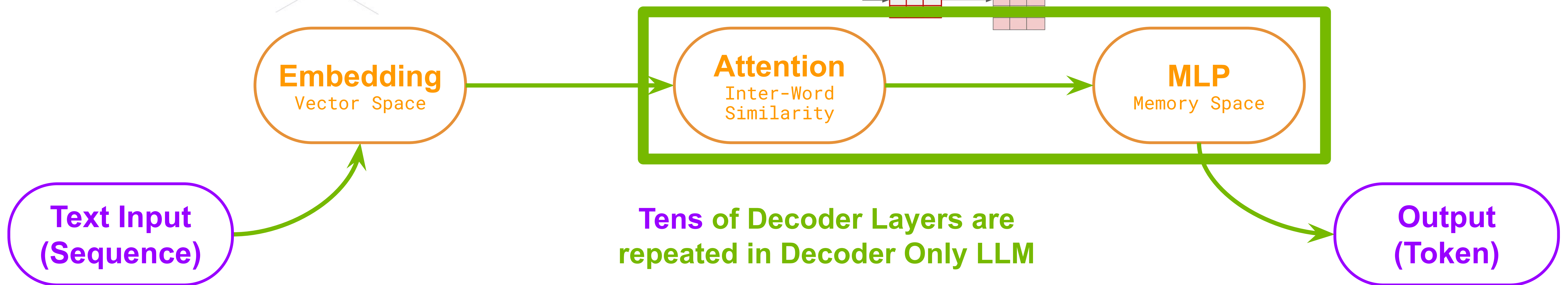
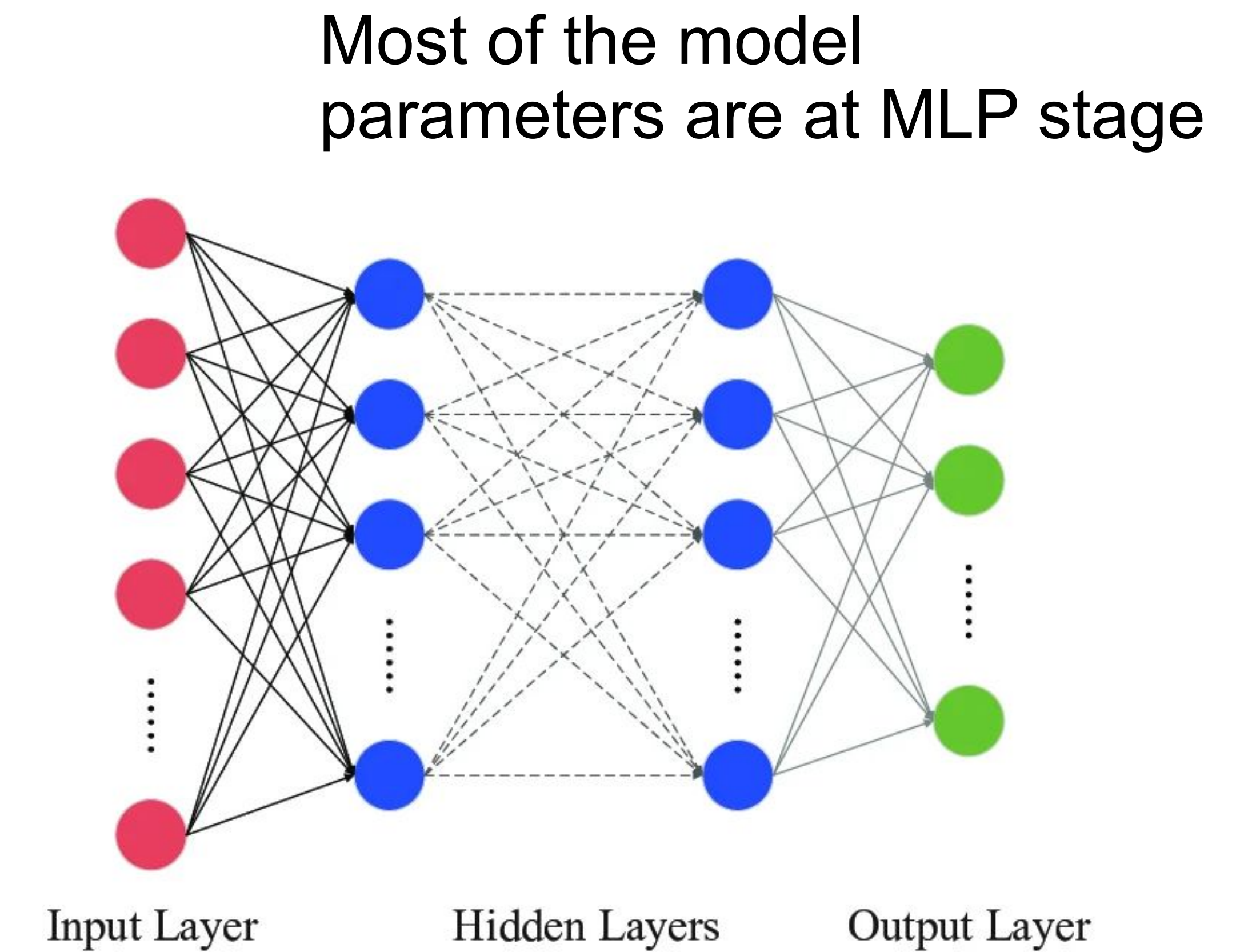
What is the Large Language Model?

Vector Space

High Dimensional Space Where Similar Words are Close and Different Words are Far

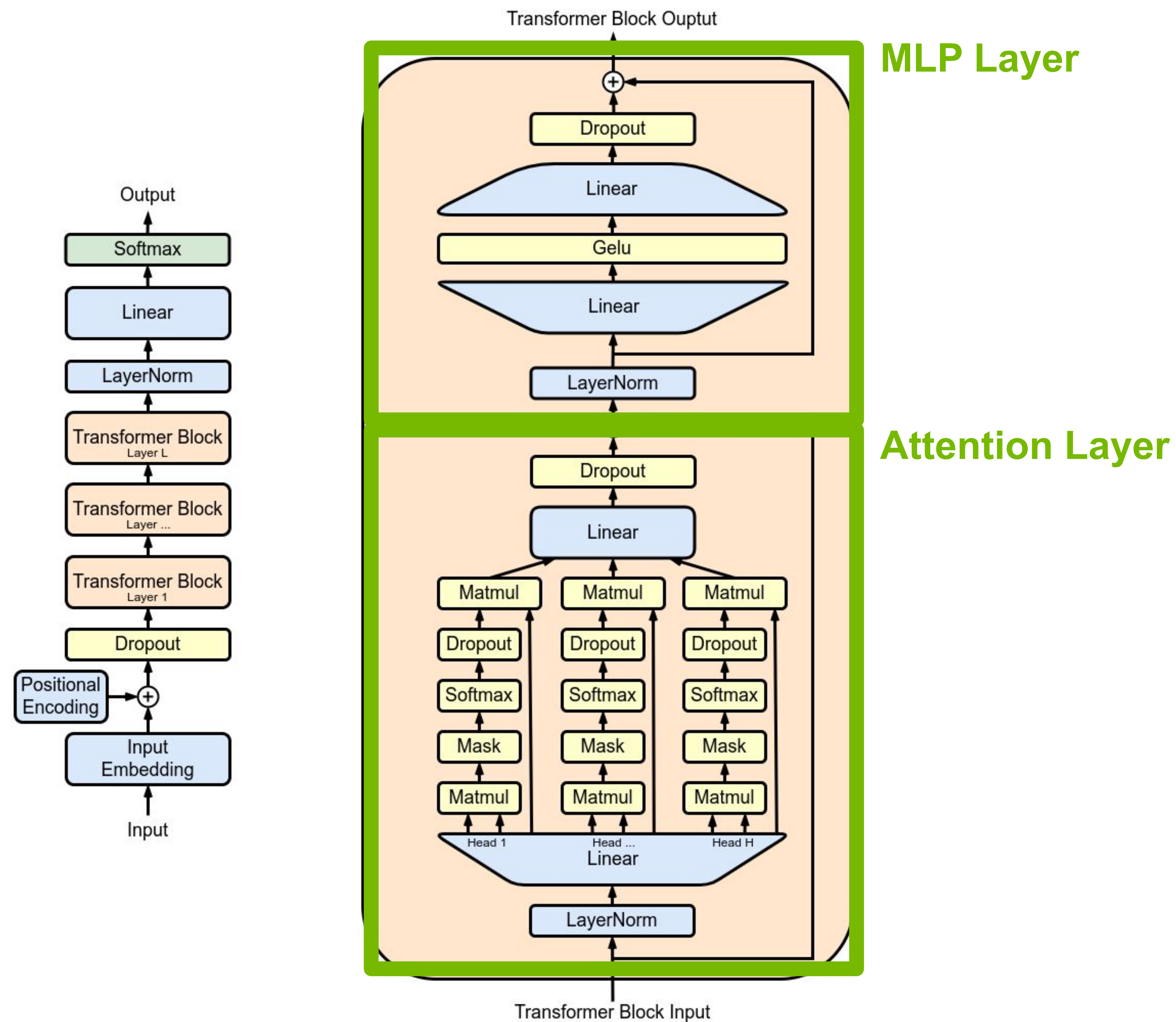


Decoder Module



Introduction to LLM

What is the Large Language Model?



Introduction to LLM

Rapid Advances in LLM Performance

*Remember the 2022?
The ChatGPT Moment!*

ChatGPT 3.5 ▾



You

What is this?



ChatGPT

This is a text-based conversational AI powered by OpenAI's GPT-3.5 model. It is designed to provide responses and engage in conversation on a wide range of topics. You can ask questions, seek information, or have a general conversation with me. Is there something specific you would like to know or discuss?



You

Chatbots have advanced significantly since I last used one



ChatGPT

Indeed, chatbot technology has made significant advancements in recent years. With the introduction of machine learning techniques, such as deep learning and neural networks,

Message ChatGPT...



ChatGPT can make mistakes. Consider checking important information.

Introduction to LLM

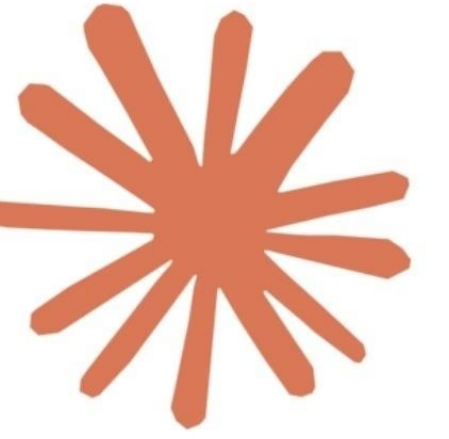
Rapid Advances in LLM Performance

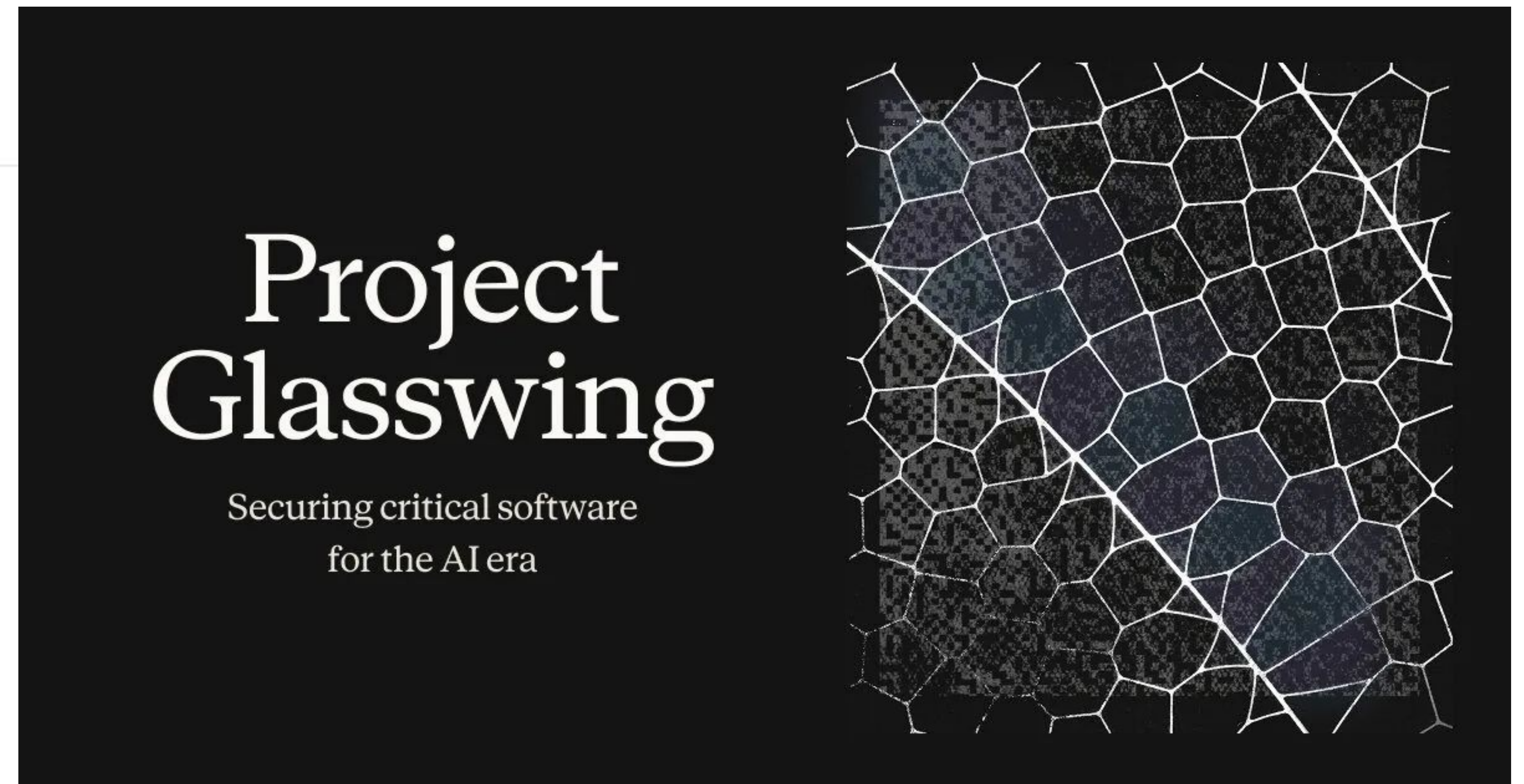


오픈AI, 차세대 AI모델 '아스트라' 예고..."수학 난제 10개 해결"

송고 2026-08-03 02:27

AI Models in 2026
→ **AI That Goes Beyond Human Capabilities**

Claude
Mythos 

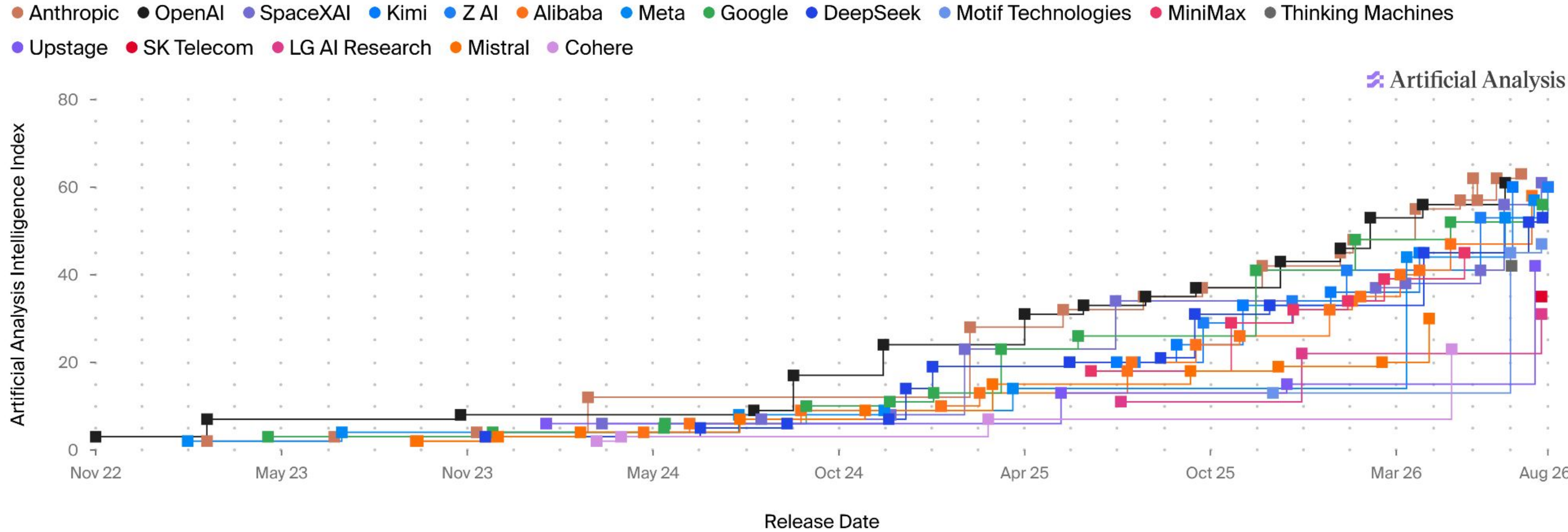


Introduction to LLM

Rapid Advances in LLM Performance

Frontier Language Model Intelligence, Over Time

Artificial Analysis Intelligence Index v4.1.1 incorporates 9 evaluations: GDPval-AA v2, τ^3 -Banking, Terminal-Bench v2.1, SciCode, Humanity's Last Exam, GPQA Diamond, CritPt, AA-Omniscience, AA-LCR



Introduction to LLM

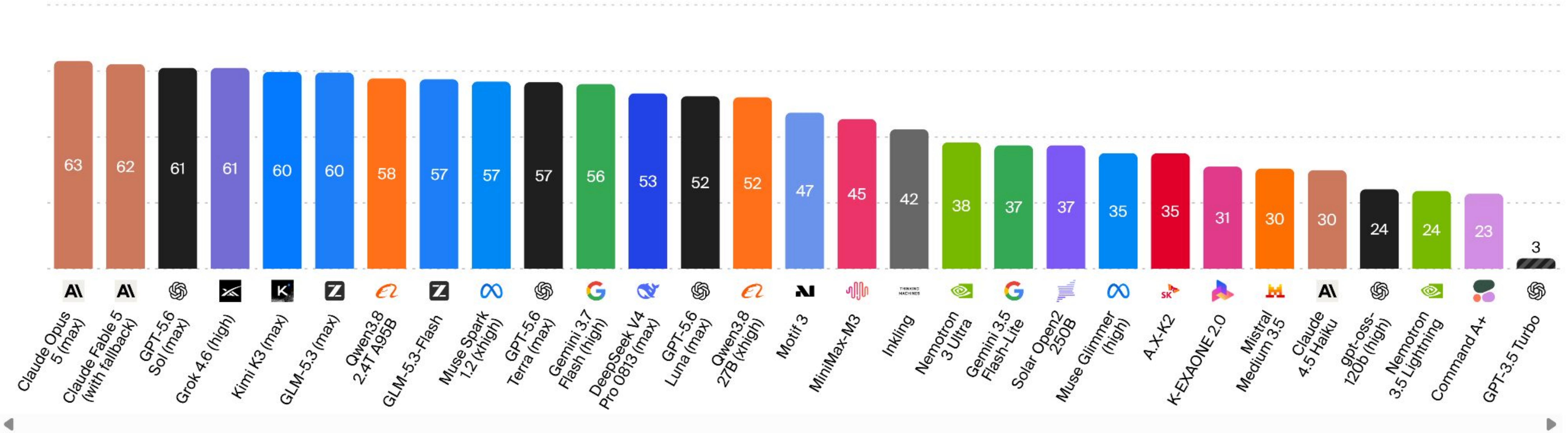
Rapid Advances in LLM Performance

Artificial Analysis Intelligence Index ↗

Artificial Analysis Intelligence Index v4.1.1 incorporates 9 evaluations: GDPval-AA v2, τ^3 -Banking, Terminal-Bench v2.1, SciCode, Humanity's Last Exam, GPQA Diamond, CritPt, AA-Omniscience, AA-LCR

▨ Estimate (independent evaluation forthcoming)

Artificial Analysis





Building RAG Agents with LLMs

Part 2. LLM Reasoning and Agentic AI

LLM Reasoning and Agentic AI

2025–2026: The Era of AI Agent

AI Models in 2026
→ *AI Agent*

🦏 OpenClaw — Personal AI Assistant



EXFOLIATE! EXFOLIATE!

BUILD **PASSING** RELEASE **V2026.2.12** DISCORD **18K ONLINE** LICENS

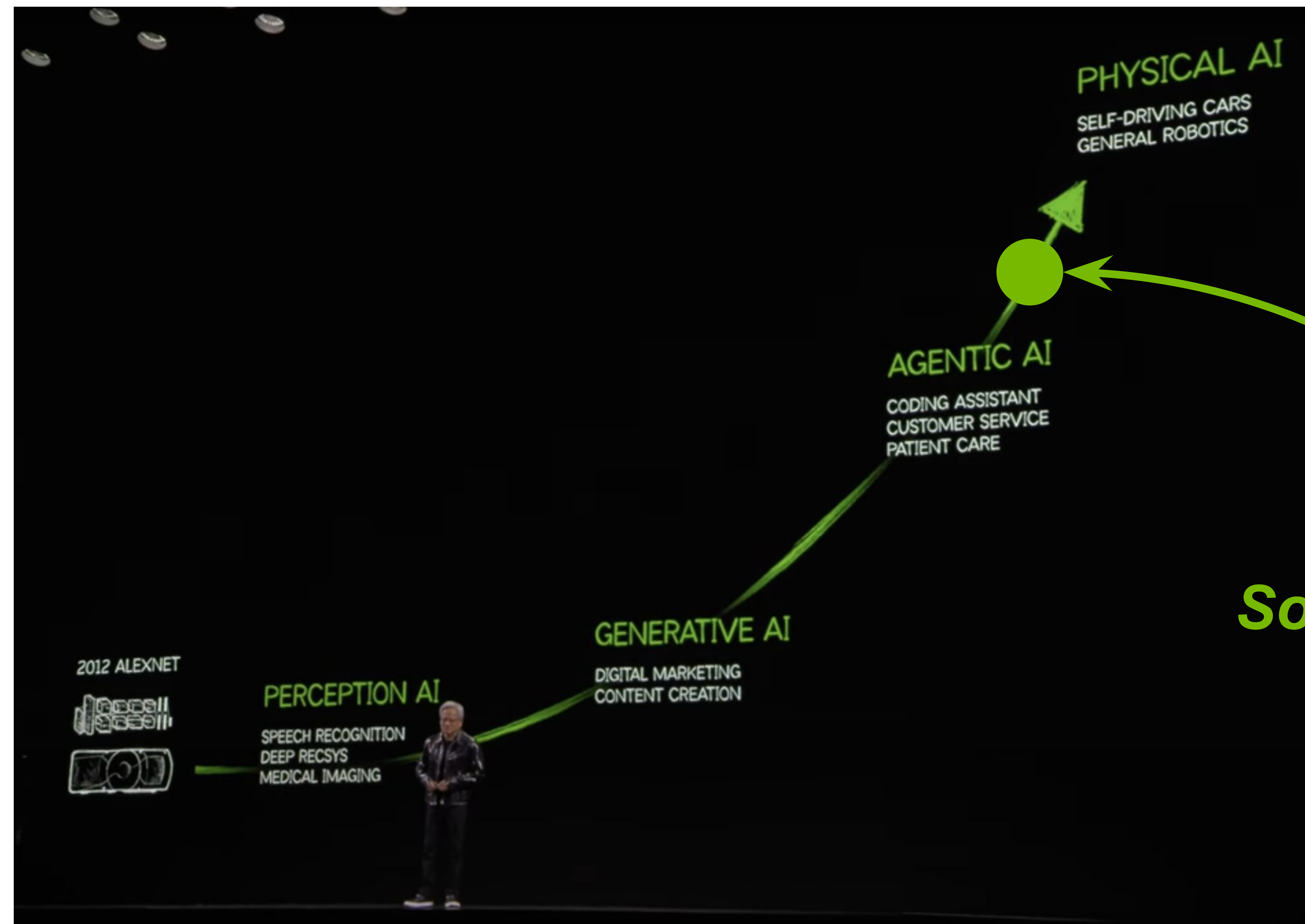
OpenClaw is a *personal AI assistant* you run on your own devices. It answers you on the channels you already use (WhatsApp, Telegram, Slack, Discord, Google Chat, Signal, iMessage, Microsoft Teams, WebChat), plus extension channels like BlueBubbles, Matrix, Zalo, and Zalo Personal. It can speak and listen on macOS/iOS/Android, and can render a live Canvas you control. The Gateway is just the control plane — the product is the assistant.

If you want a personal, single-user assistant that feels local, fast, and always-on, this is it.



LLM Reasoning and Agentic AI

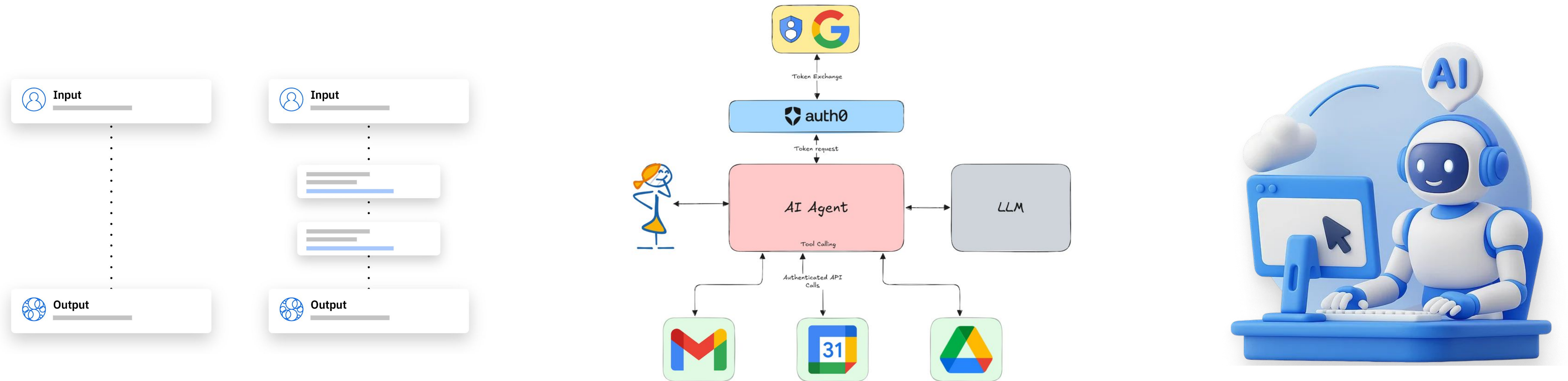
2025–2026: The Era of AI Agent



Somewhere in here

LLM Reasoning and Agentic AI

Technical Foundations of AI Agents: Reasoning & External Interaction



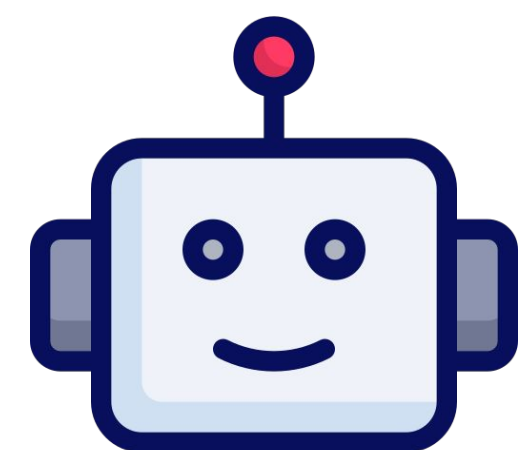
Chain-of-Thought → Function Calling → Agentic AI

LLM Reasoning and Agentic AI

Technical Foundations of AI Agents: Reasoning & External Interaction

<|system|> You are a helpful assistant. When solving a problem, think step by step before giving the final answer.

<|user|> 한 상자에 연필이 12자루 있습니다. 이런 상자가 5개 있다면 연필은 모두 몇 자루인가요?

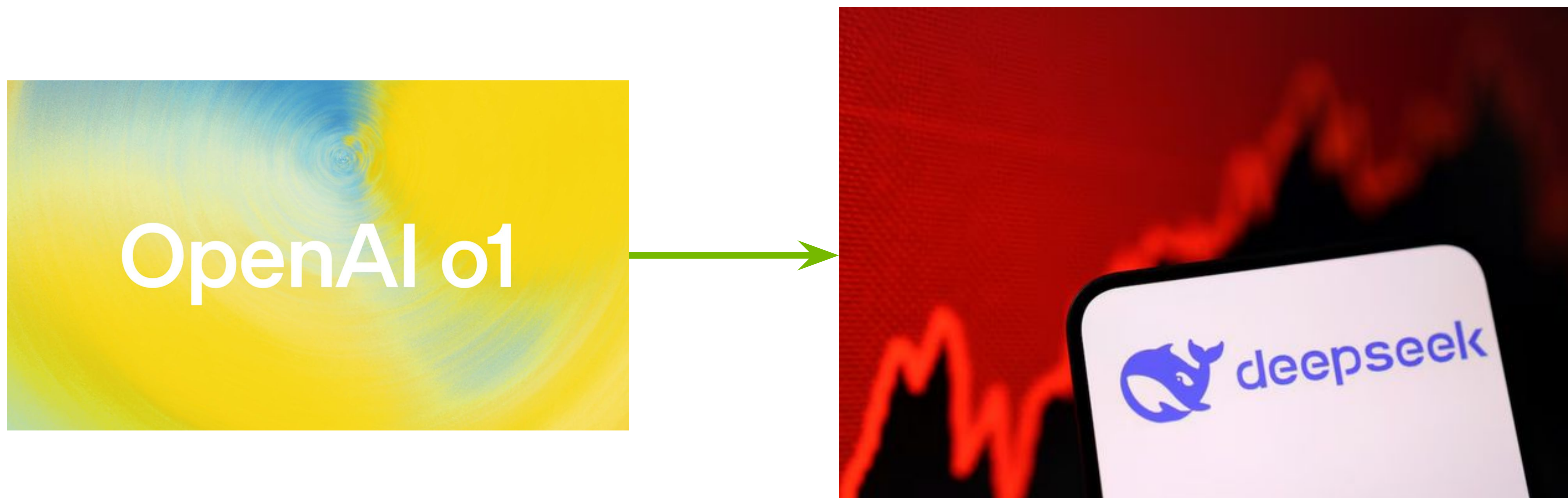


<|assistant|> <think> I need to determine the total number of pencils. Each box contains 12 pencils. There are 5 boxes. Total pencils = $12 \times 5 = 60$. So there are 60 pencils in total. </think> 총 60자루의 연필이 있습니다.

1st Evolution: Enhancing Reasoning Capabilities through CoT

LLM Reasoning and Agentic AI

Technical Foundations of AI Agents: Reasoning & External Interaction



Remember the Deepseek Shock?

→ in the Academia Field, Deepseek Moment

Academic Reproduction of Reasoning via RL

LLM Reasoning and Agentic AI

Technical Foundations of AI Agents: Reasoning & External Interaction



DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning

DeepSeek-AI
research@deepseek.com

DeepSeek-R1: How CoT Emerges?

Abstract

General reasoning represents a long-standing and formidable challenge in artificial intelligence. Recent breakthroughs, exemplified by large language models (LLMs) (Brown et al., 2020; OpenAI, 2023) and chain-of-thought prompting (Wei et al., 2022b), have achieved considerable success on foundational reasoning tasks. However, this success is heavily contingent upon extensive human-annotated demonstrations, and models' capabilities are still insufficient for more complex problems. Here we show that the reasoning abilities of LLMs can be incentivized through pure reinforcement learning (RL), obviating the need for human-labeled reasoning trajectories. The proposed RL framework facilitates the emergent development of advanced reasoning patterns, such as self-reflection, verification, and dynamic strategy adaptation. Consequently, the trained model achieves superior performance on verifiable tasks such as mathematics, coding competitions, and STEM fields, surpassing its counterparts trained via conventional supervised learning on human demonstrations. Moreover, the emergent reasoning patterns exhibited by these large-scale models can be systematically harnessed to guide and enhance the reasoning capabilities of smaller models.

.12948v2 [cs.CL] 4 Jan 2026

LLM Reasoning and Agentic AI

Technical Foundations of AI Agents: Reasoning & External Interaction

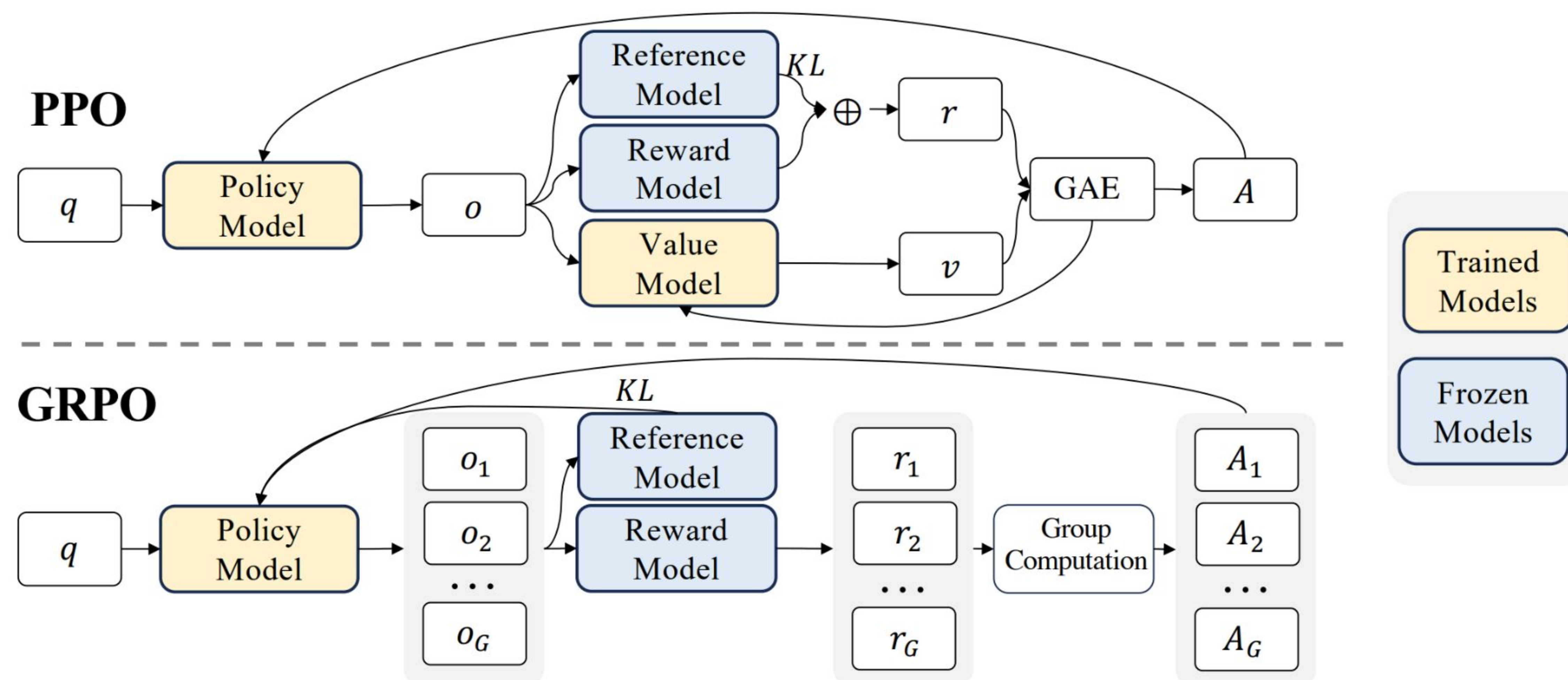


Figure 4 | Demonstration of PPO and our GRPO. GRPO foregoes the value model, instead estimating the baseline from group scores, significantly reducing training resources.

LLM Reasoning and Agentic AI

Technical Foundations of AI Agents: Reasoning & External Interaction

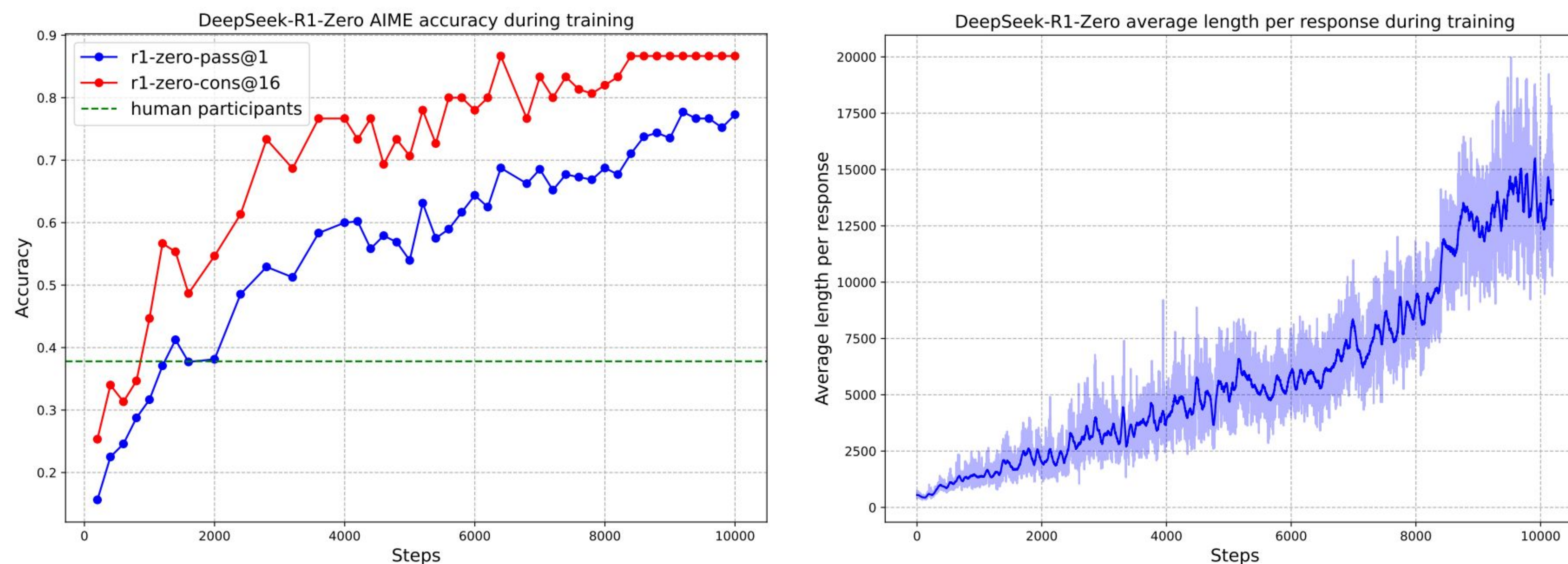


Figure 1 | (a) AIME accuracy of DeepSeek-R1-Zero during training. AIME takes a mathematical problem as input and a number as output, illustrated in Table 32. Pass@1 and Cons@16 are described in Supplementary D.1. The baseline is the average score achieved by human participants in the AIME competition. (b) The average response length of DeepSeek-R1-Zero on the training set during the RL process. DeepSeek-R1-Zero naturally learns to solve reasoning tasks with more thinking time. Note that a training step refers to a single policy update operation.

LLM Reasoning and Agentic AI

Technical Foundations of AI Agents: Reasoning & External Interaction

Table 2 | An interesting “aha moment” of an intermediate version of DeepSeek-R1-Zero. The model learns to rethink using an anthropomorphic tone. This is also an aha moment for us, allowing us to witness the power and beauty of reinforcement learning.

Question: If $a > 1$, then the sum of the real solutions of $\sqrt{a - \sqrt{a + x}} = x$ is equal to

Response: <think>

To solve the equation $\sqrt{a - \sqrt{a + x}} = x$, let's start by squaring both ...

$$\left(\sqrt{a - \sqrt{a + x}}\right)^2 = x^2 \implies a - \sqrt{a + x} = x^2.$$

Rearrange to isolate the inner square root term:

$$(a - x^2)^2 = a + x \implies a^2 - 2ax^2 + (x^2)^2 = a + x \implies x^4 - 2ax^2 - x + (a^2 - a) = 0$$

...

Wait, wait. Wait. That's an aha moment I can flag here.

Let's reevaluate this step-by-step to identify if the correct sum can be ...

We started with the equation:

$$\sqrt{a - \sqrt{a + x}} = x$$

First, let's square both sides:

$$a - \sqrt{a + x} = x^2 \implies \sqrt{a + x} = a - x^2$$

Next, I could square both sides again, treating the equation: ...

...

LLM Reasoning and Agentic AI

Technical Foundations of AI Agents: Reasoning & External Interaction

B.4.4. Training Cost

Regarding our research on DeepSeek-R1, we utilized the A100 GPUs to prepare for the experiments with a smaller model (30B parameters). The results from this smaller model have been promising, which has allowed us to confidently scale up to 660B R1-Zero and R1. For the training of DeepSeek-R1-Zero, we employed 64*8 H800 GPUs, and the process required approximately 198 hours. Additionally, during the training phase of DeepSeek-R1, we utilized the same 64*8 H800 GPUs, completing the process in about 4 days, or roughly 80 hours. To create the SFT datasets, we use 5K GPU hours. The details are shown in Table 7.

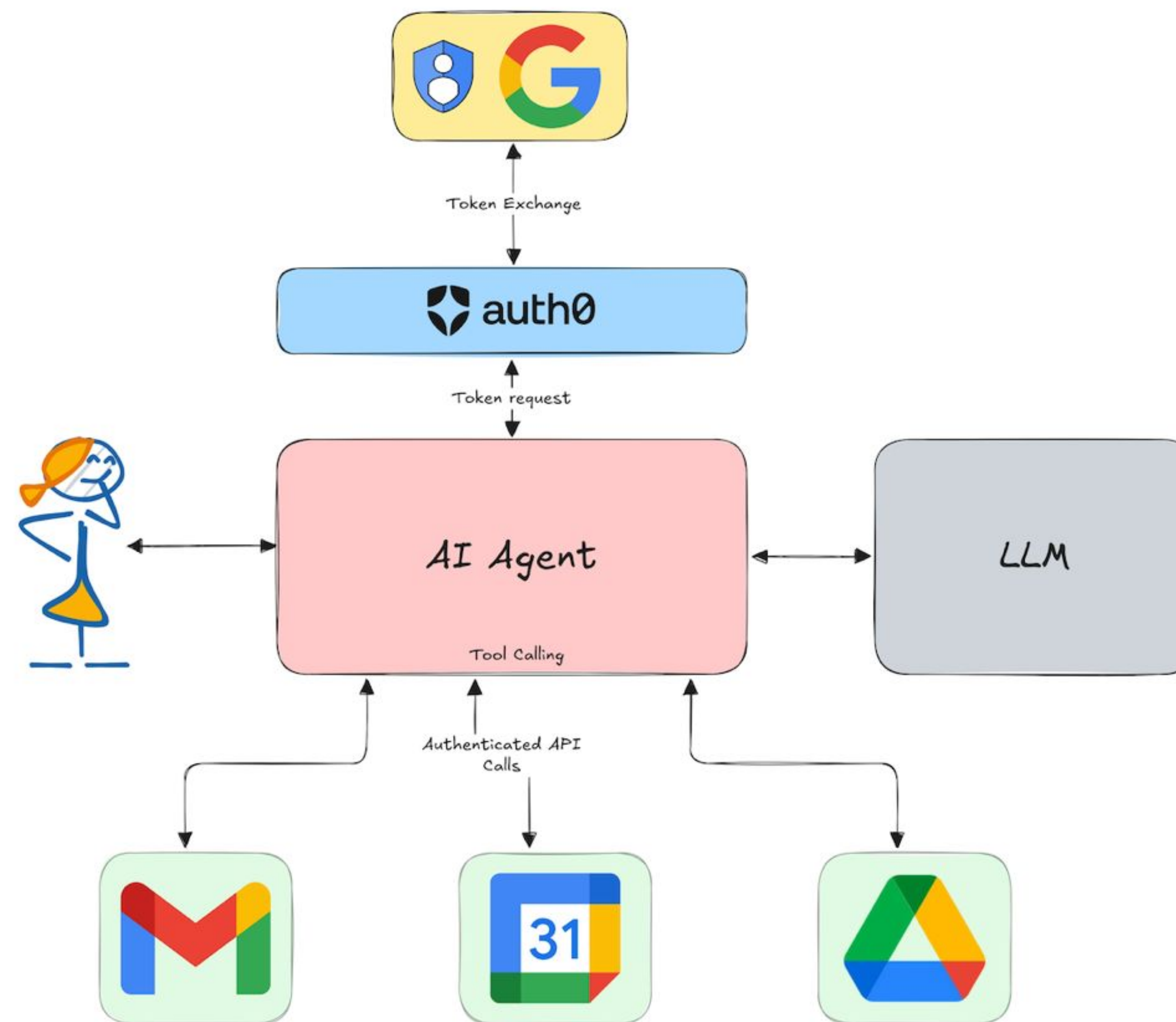
Training a GPT-level AI for approximately \$5.6 million (~~~\$8 billion~~)

Model	AIME 2024		MATH-500	GPQA Diamond	LiveCode Bench	CodeForces
	pass@1	cons@64	pass@1	pass@1	pass@1	rating
OpenAI-o1-mini	63.6	80.0	90.0	60.0	53.8	1820
OpenAI-o1-0912	74.4	83.3	94.8	77.3	63.4	1843
DeepSeek-R1-Zero	71.0	86.7	95.9	73.3	50.0	1444

Table 2 | Comparison of DeepSeek-R1-Zero and OpenAI o1 models on reasoning-related benchmarks.

LLM Reasoning and Agentic AI

Technical Foundations of AI Agents: Reasoning & External Interaction



2nd Evolution: Extending Capabilities through Tool Use

LLM Reasoning and Agentic AI

Technical Foundations of AI Agents: Reasoning & External Interaction

<|start|>

Tools

You have access to the following tool:

get_weather(location: string)

→ Returns the current weather for a given location.

Tool Call Format

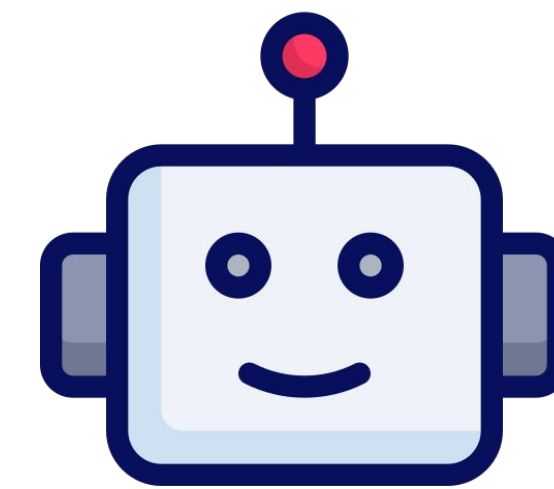
<|tool_call|>

{"name": "<function_name>", "arguments": {...}}

<|end|>

User

What's the weather in Seoul today?



<|assistant|>

<|tool_call|>

{"name": "get_weather", "arguments": {"location": "Seoul"}}

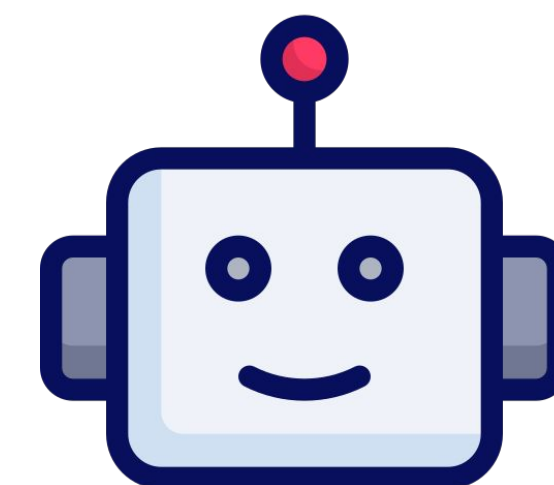
<|end|>



<|tool_result|>

{"temperature": 27, "condition": "Sunny"}

<|end|>



<|assistant|>

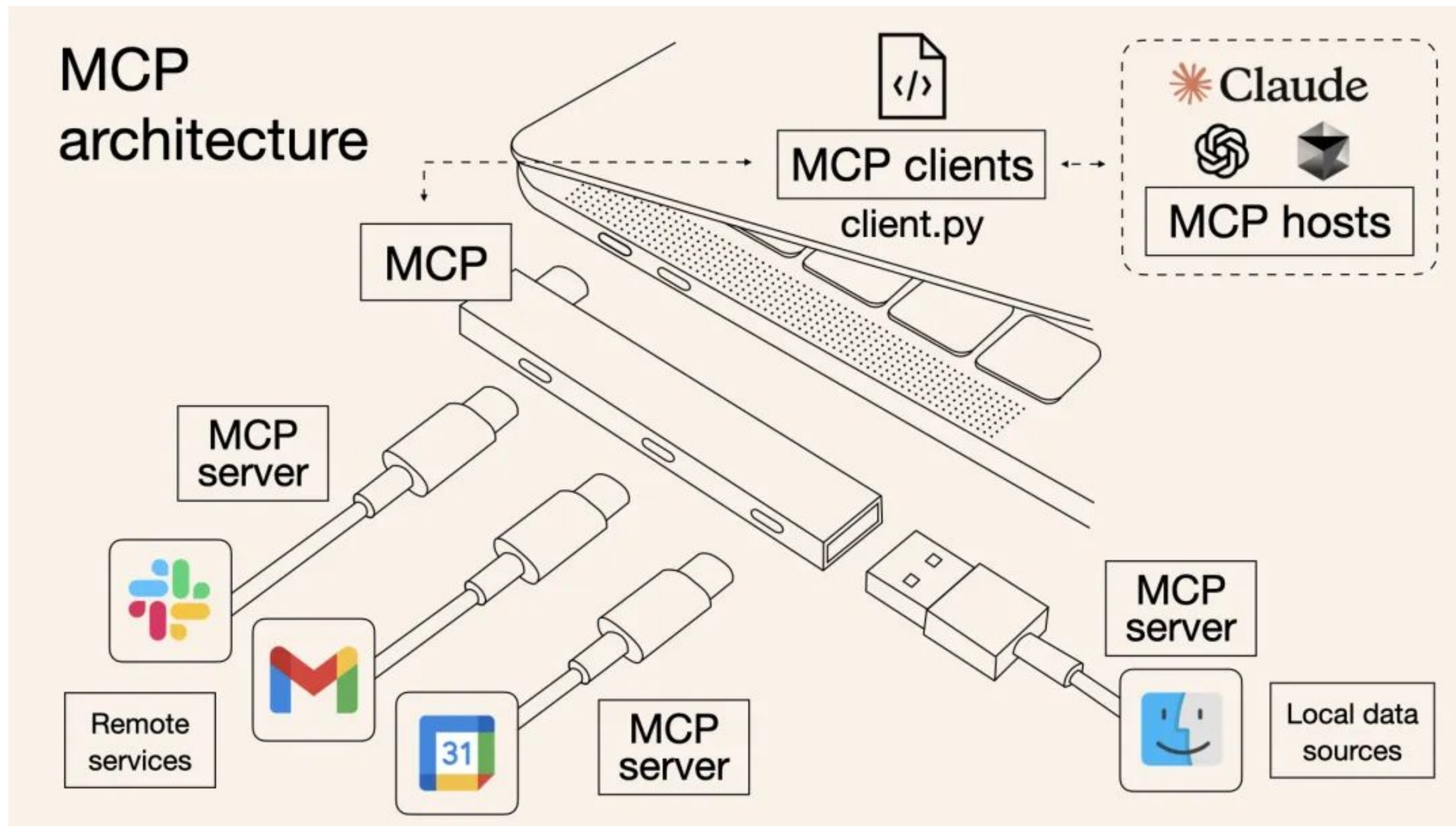
It's 27°C and sunny in Seoul today.

<|end|>

2nd Evolution: Extending Capabilities through Tool Use

LLM Reasoning and Agentic AI

Technical Foundations of AI Agents: Reasoning & External Interaction



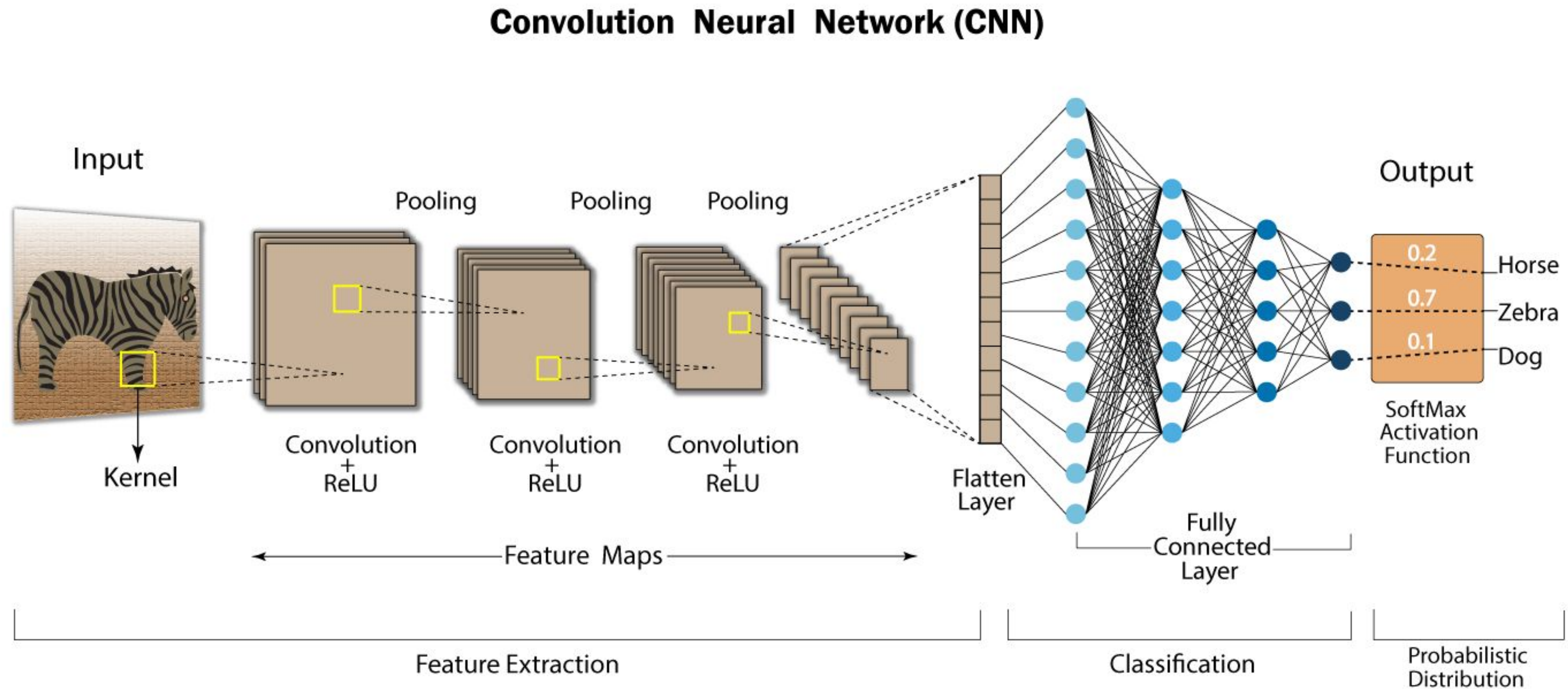


Building RAG Agents with LLMs

Part 3. LLM Training

LLM Training

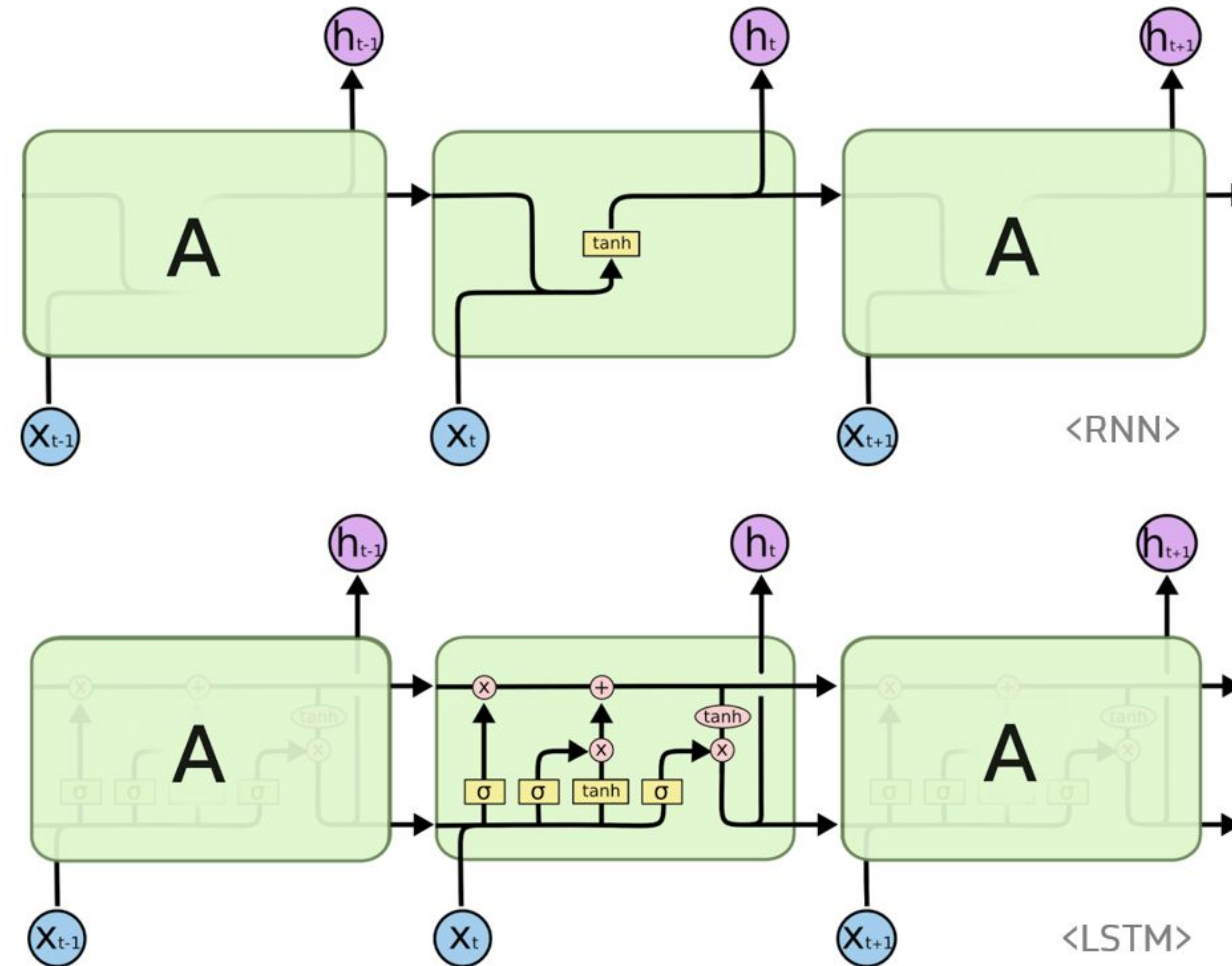
Why Transformer is Powerful Architecture?



Inductive Bias : Nearby pixels are more likely to be related.
→ Image Prediction Task

LLM Training

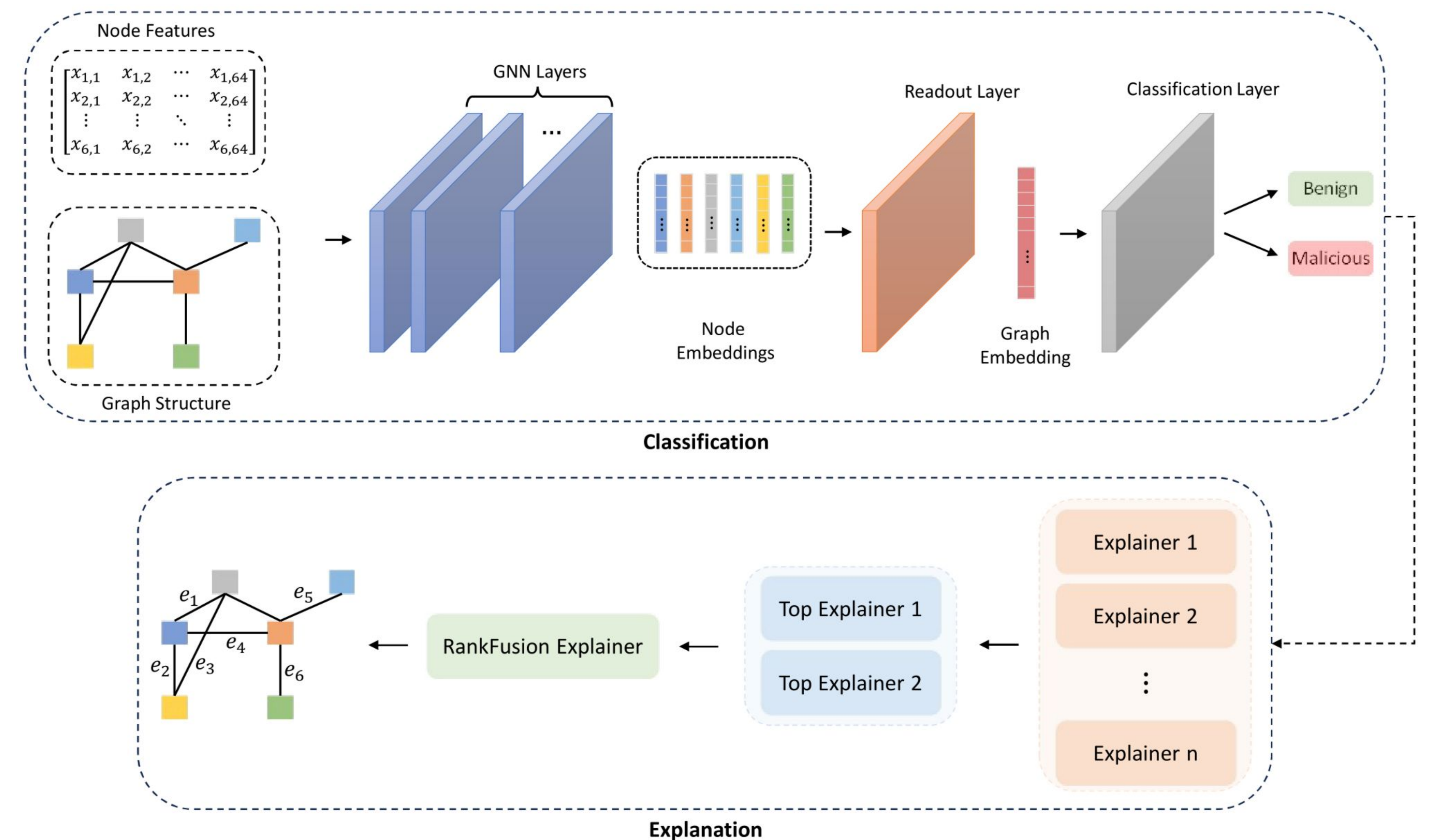
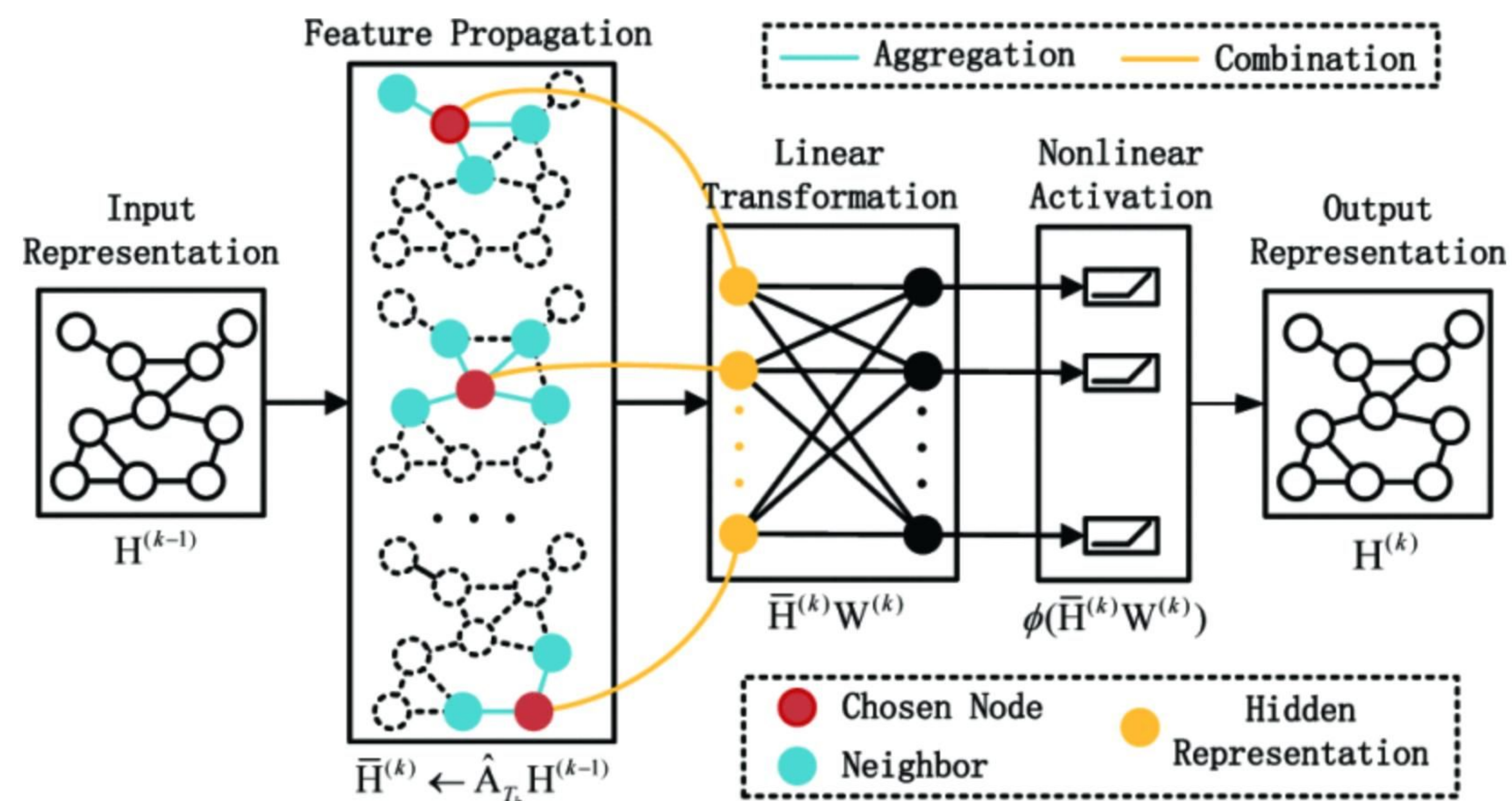
Why Transformer is Powerful Architecture?



Inductive Bias : The current state depends on previous states.
→ **Sequence Prediction Task**

LLM Training

Why Transformer is Powerful Architecture?



Inductive Bias : Connected nodes are more likely to interact.

→ Graph Prediction Task

Figure 1: A Sankey diagram illustrating word co-occurrence between two sentences. The left sentence is "It is in this spirit that a majority of American governments have passed new laws since 2009 making the registration or voting process more difficult." and the right sentence is "It is in this spirit that a majority of American governments have passed new laws since 2009 making the registration or voting process more difficult." The diagram shows the flow of words from the left sentence to the right sentence, with the word "making" being the most prominent co-occurrence.



LLM Training

Massive Computational and Data Resources for Model Training



LLM Training

Massive Computational and Data Resources for Model Training



Inside an AI start-up's plan to scan and dispose of millions of books

Court filings reveal how AI companies raced to obtain more books to feed chatbots, including by buying, scanning and disposing of millions of titles.

January 27, 2026

Massive Data
to Model Training



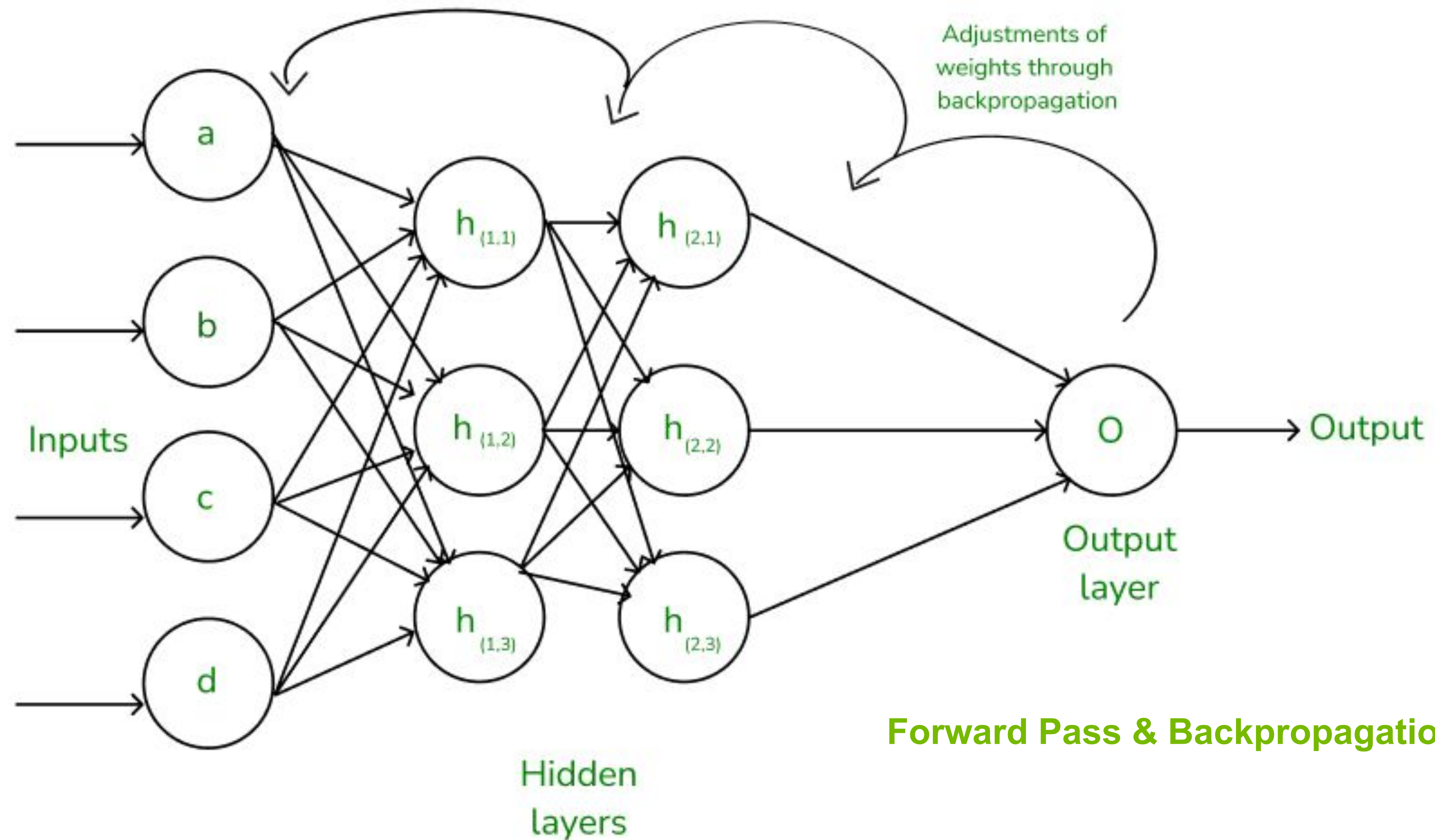
LLM Training

Massive Computational and Data Resources for Model Training



LLM Training

How are LLM models trained?



Forward Pass & Backpropagation

LLM Training

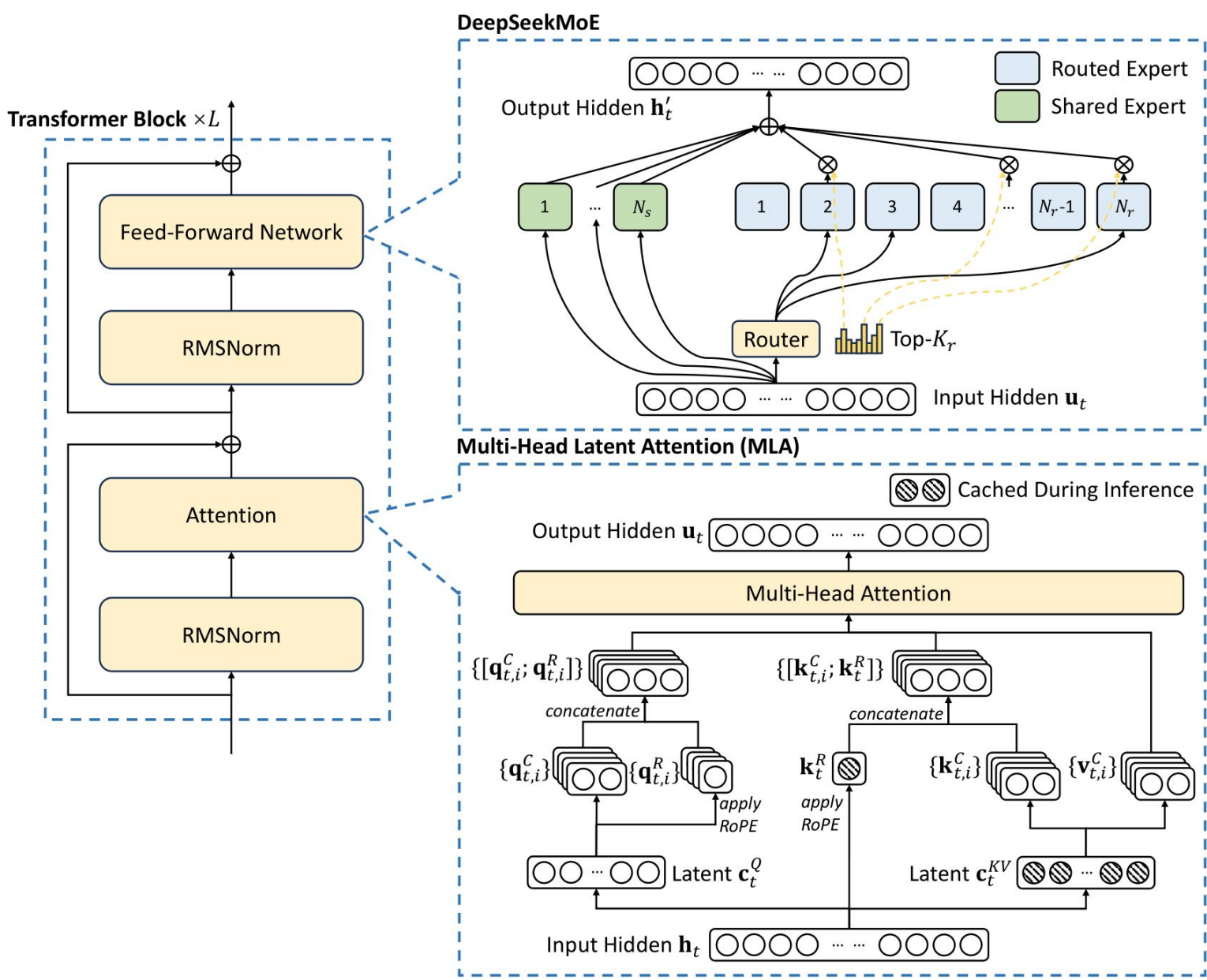
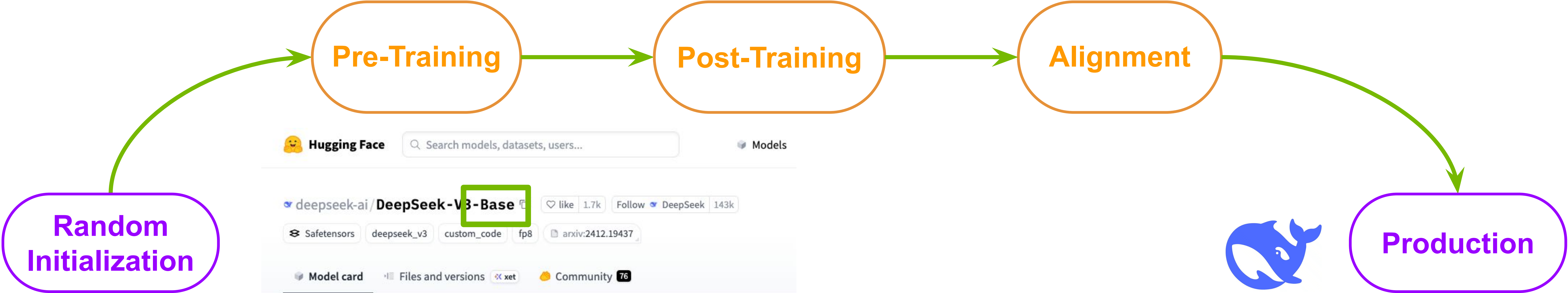
How are LLM models trained?

Next-Token Prediction Problem
with ~ Trillion Corpus

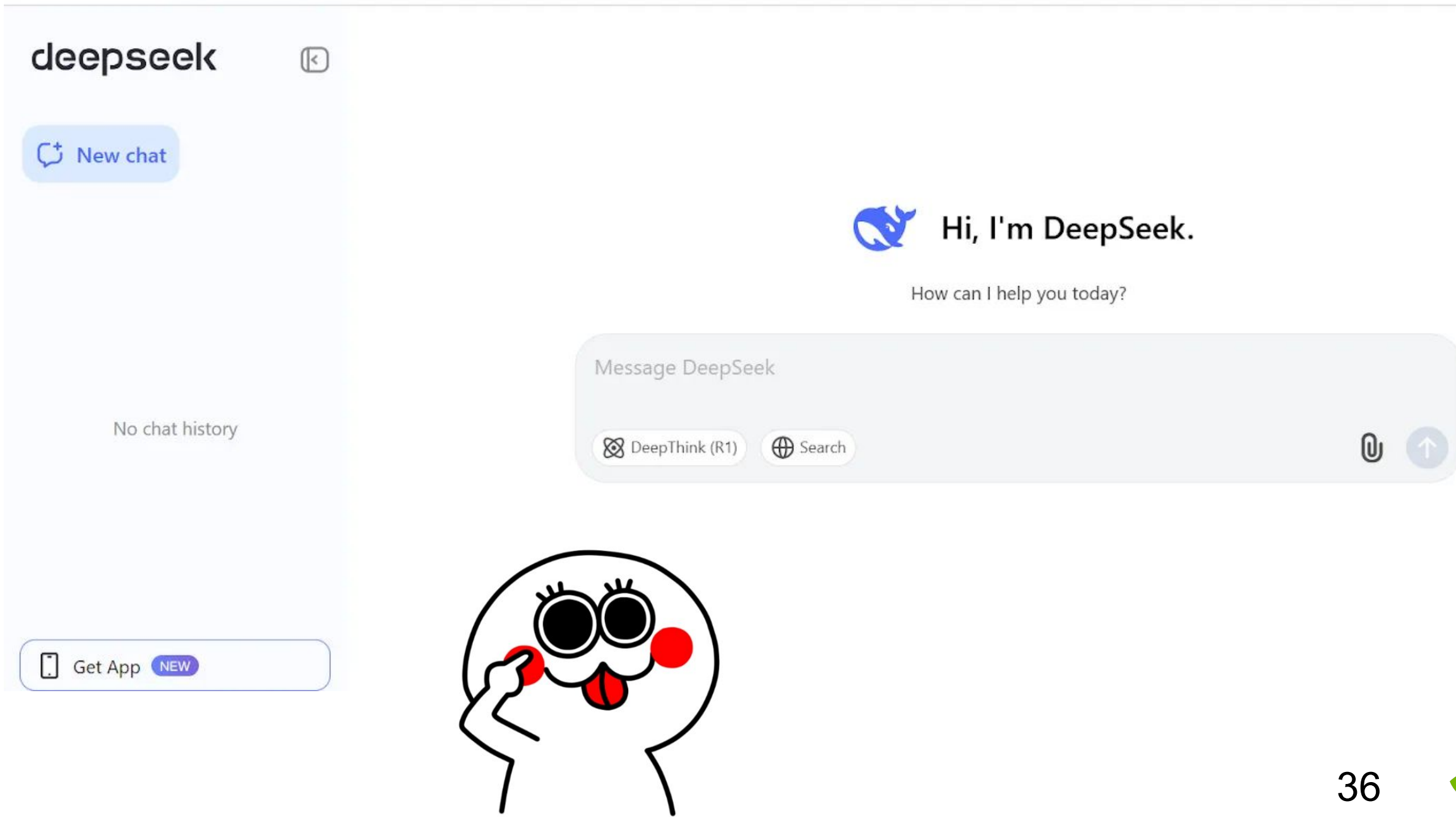
Code, Math...
Task Specific Knowledge



Guardrail, Human Preference ...



Strong Backbone
Architecture with
randomized parameters

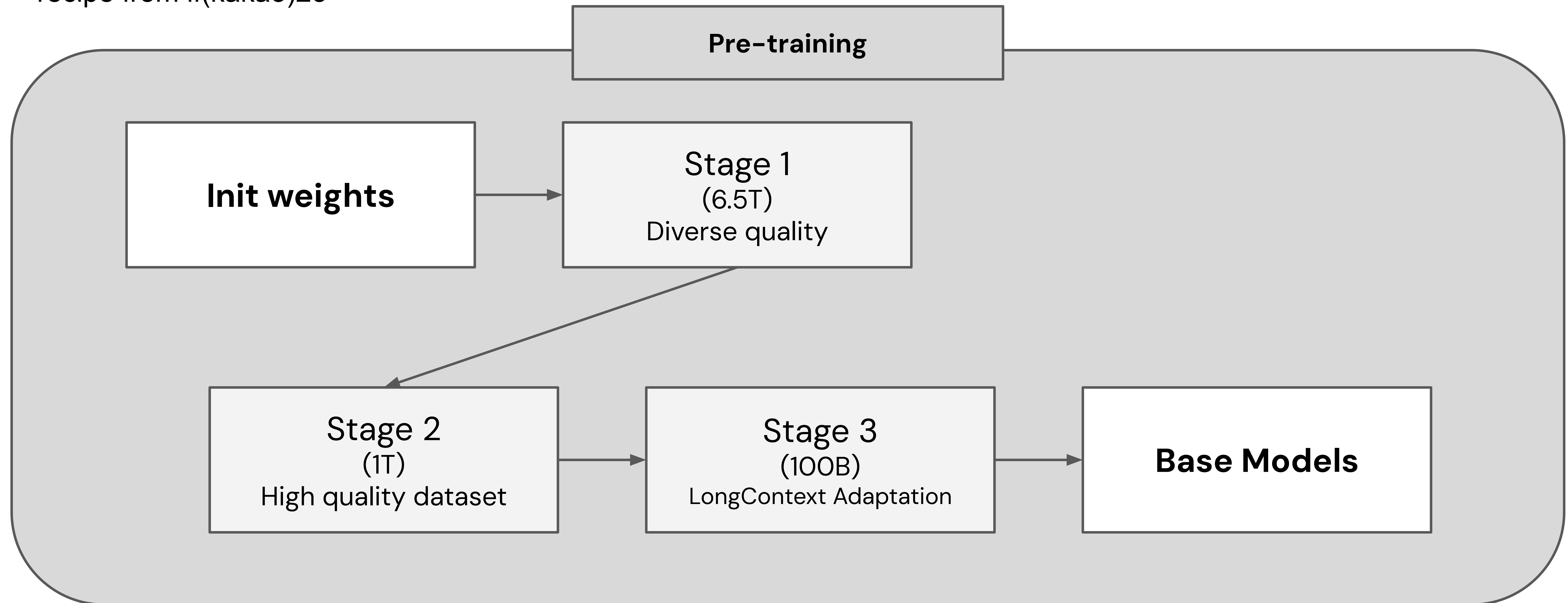


LLM Training

How are LLM models trained?

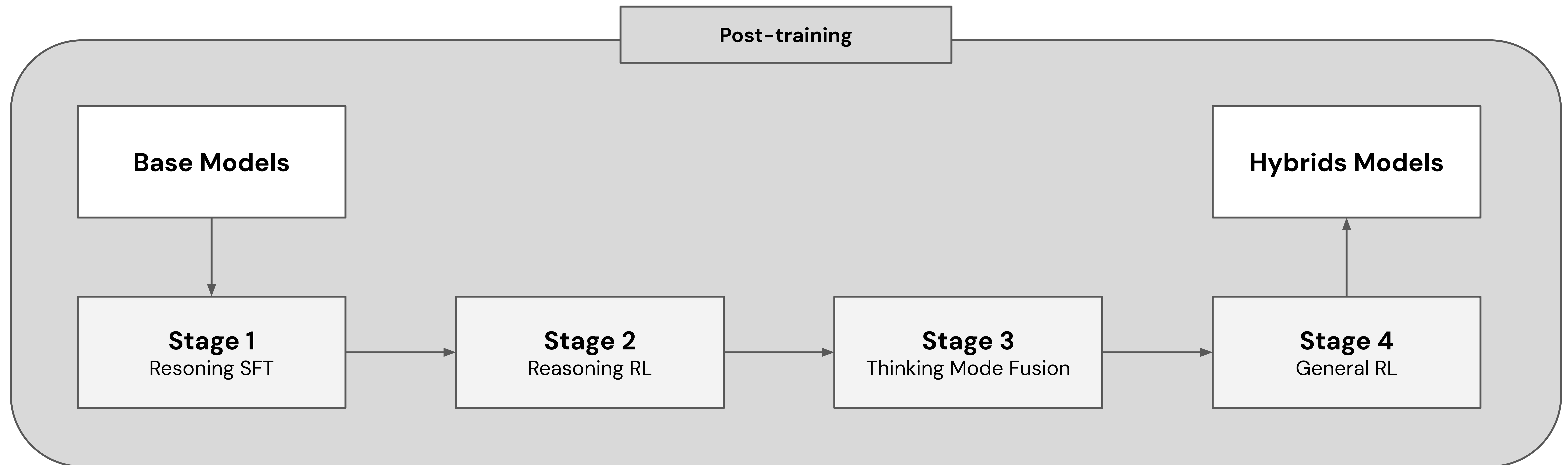


Kanana(Kakao) model training
recipe from if(kakao)25



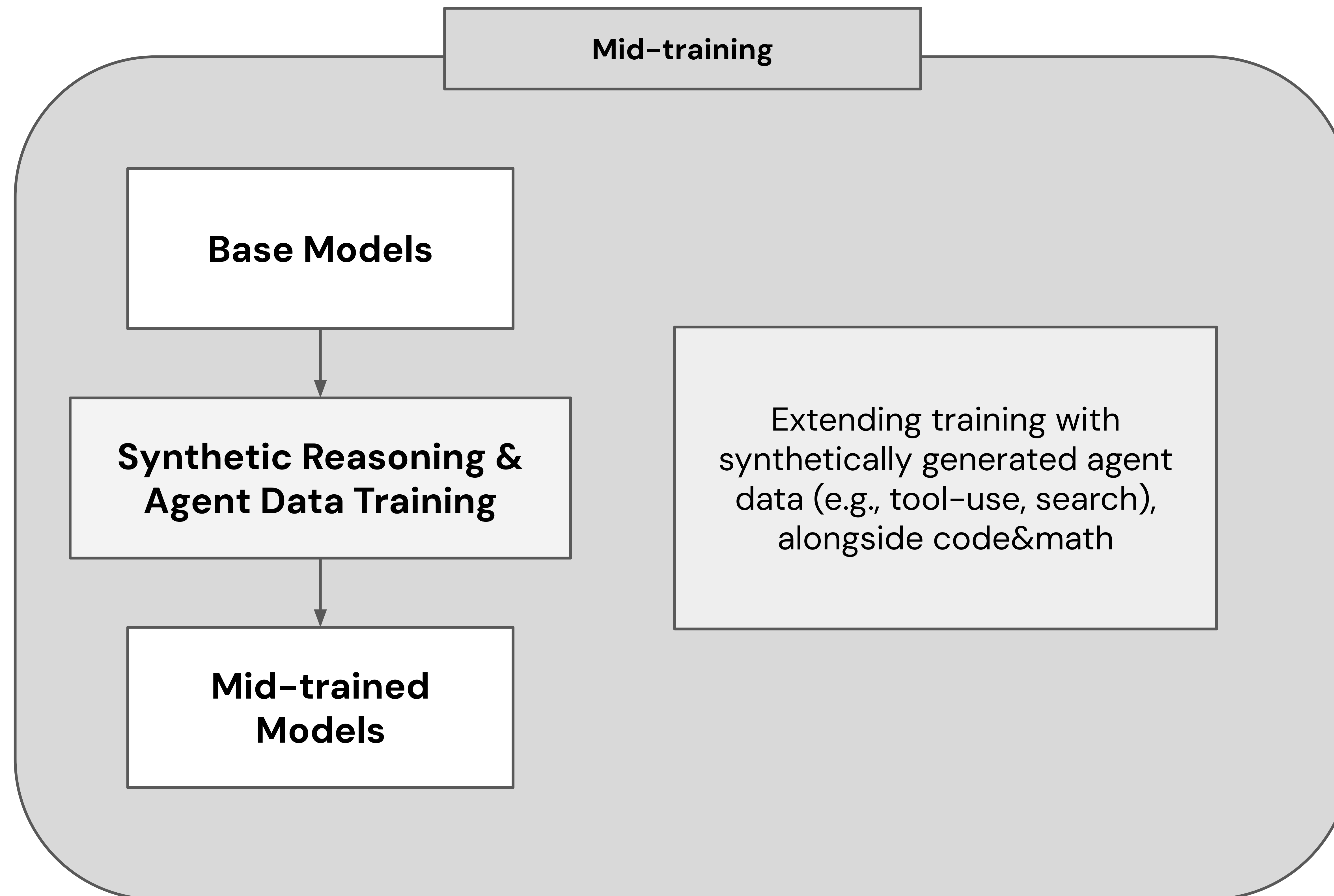
LLM Training

How are LLM models trained?



LLM Training

How are LLM models trained?



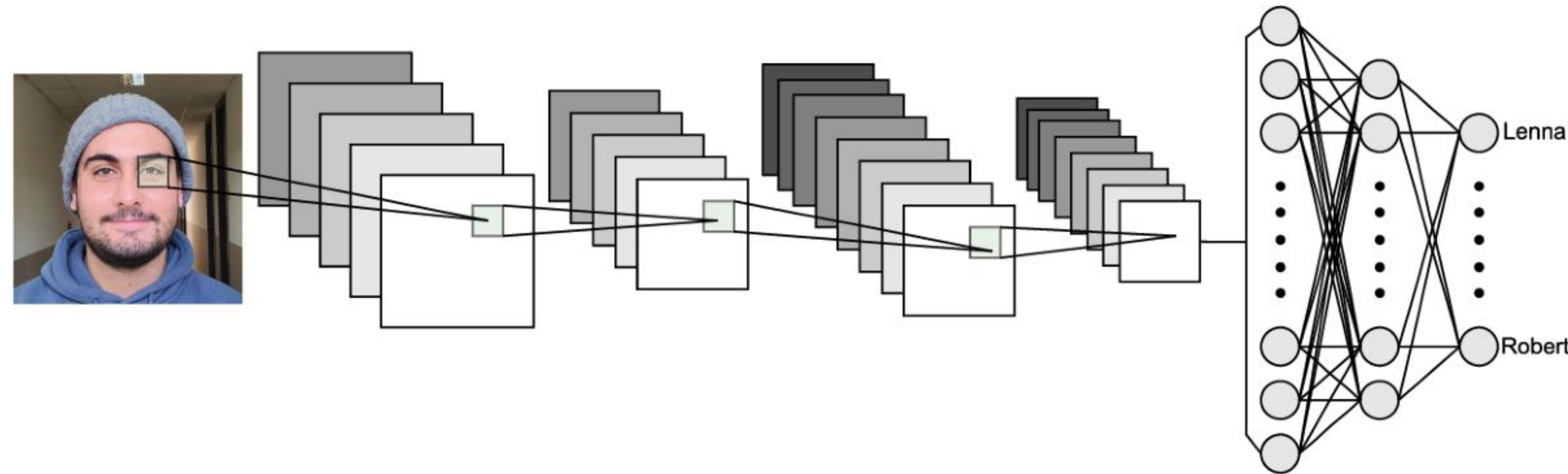


Building RAG Agents with LLMs

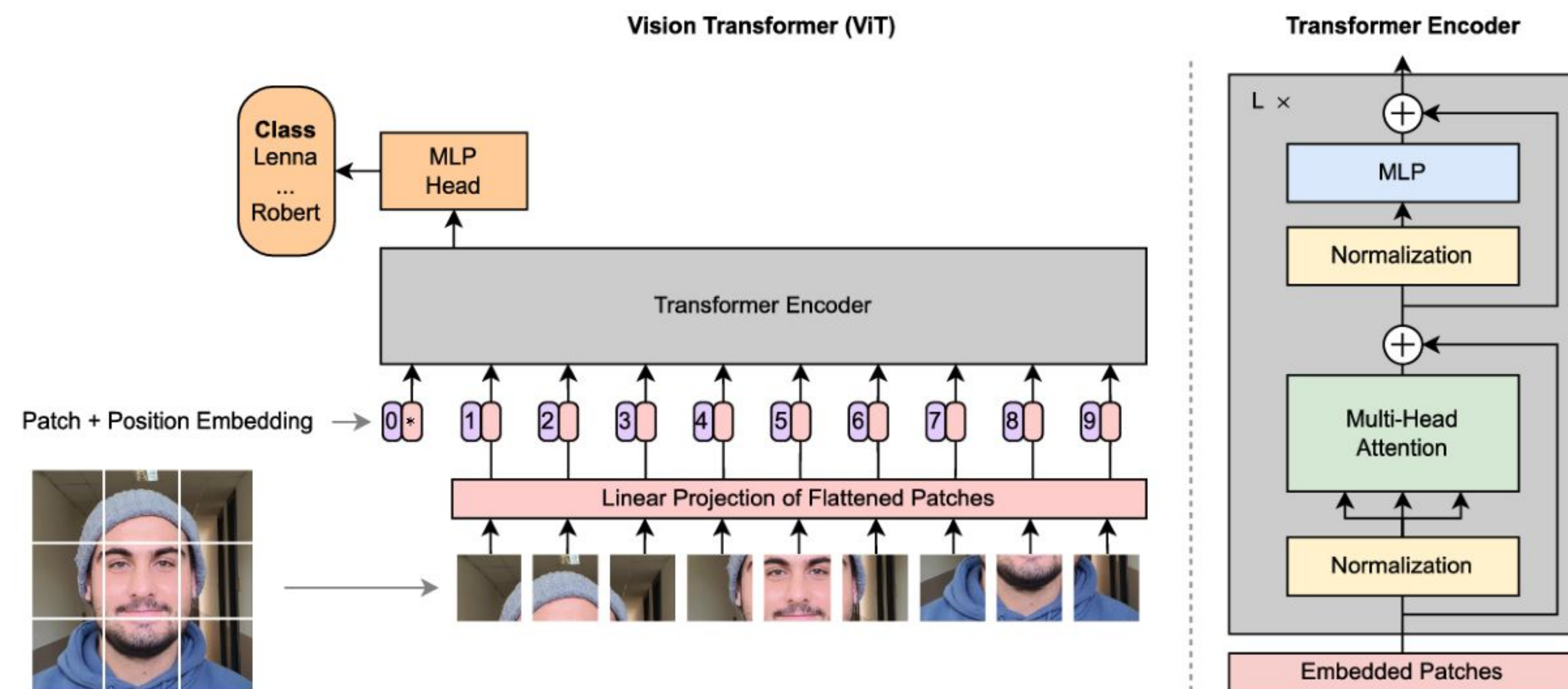
Part 4. Evolution of AI Models

Evolution of AI Models

Expansion into Multimodality and Omni-Modality



(a) Common CNN architecture



(b) Vision Transformer architecture

Evolution of AI Models

Expansion into Multimodality and Omni-Modality

NExT-GPT: Any-to-Any Multimodal LLM

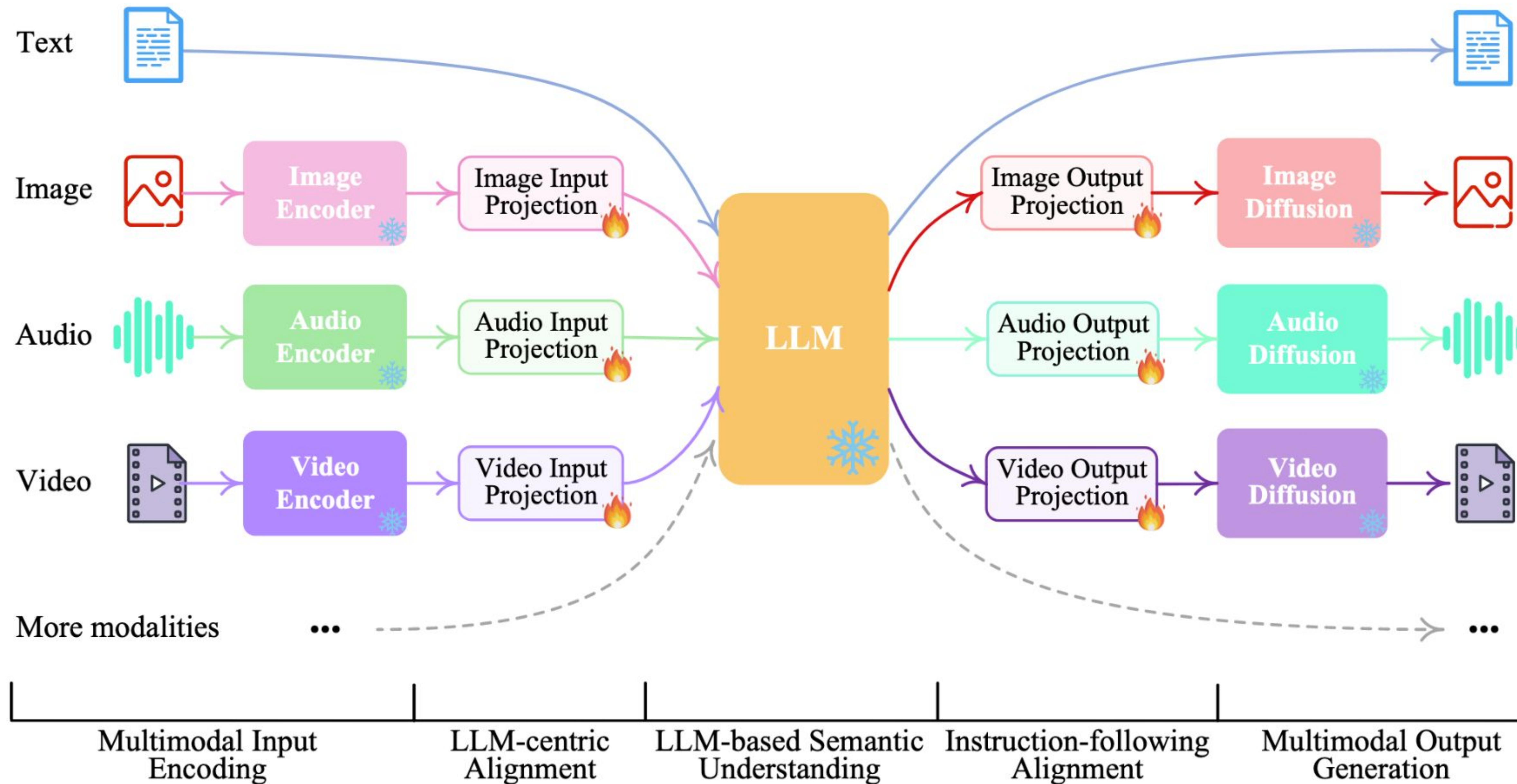
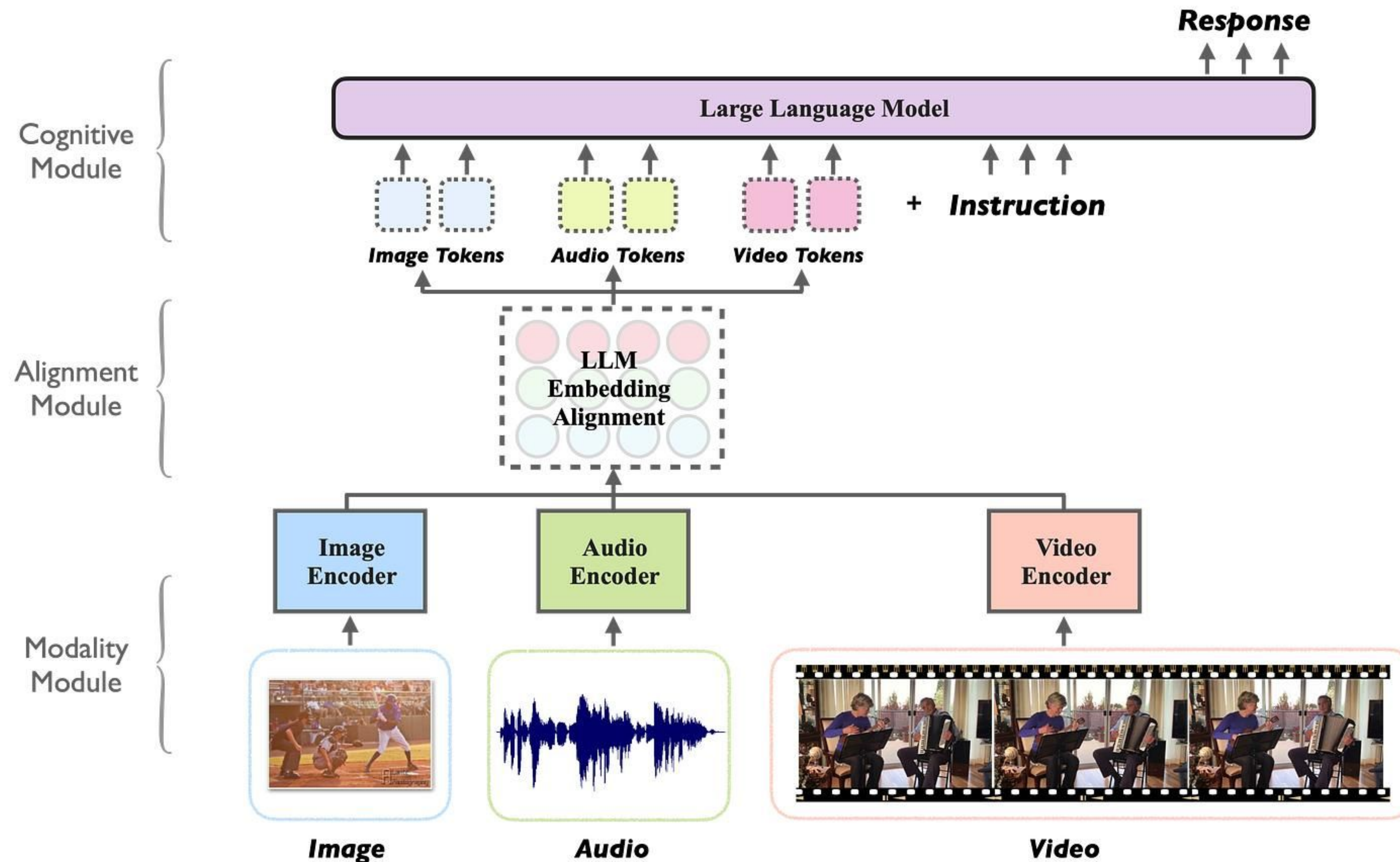


Figure 1. By connecting LLM with multimodal adaptors and diffusion decoders, NExT-GPT achieves universal multimodal understanding and any-to-any modality input and output. ❄️ and 🔥 represent the frozen and trainable modules, respectively.

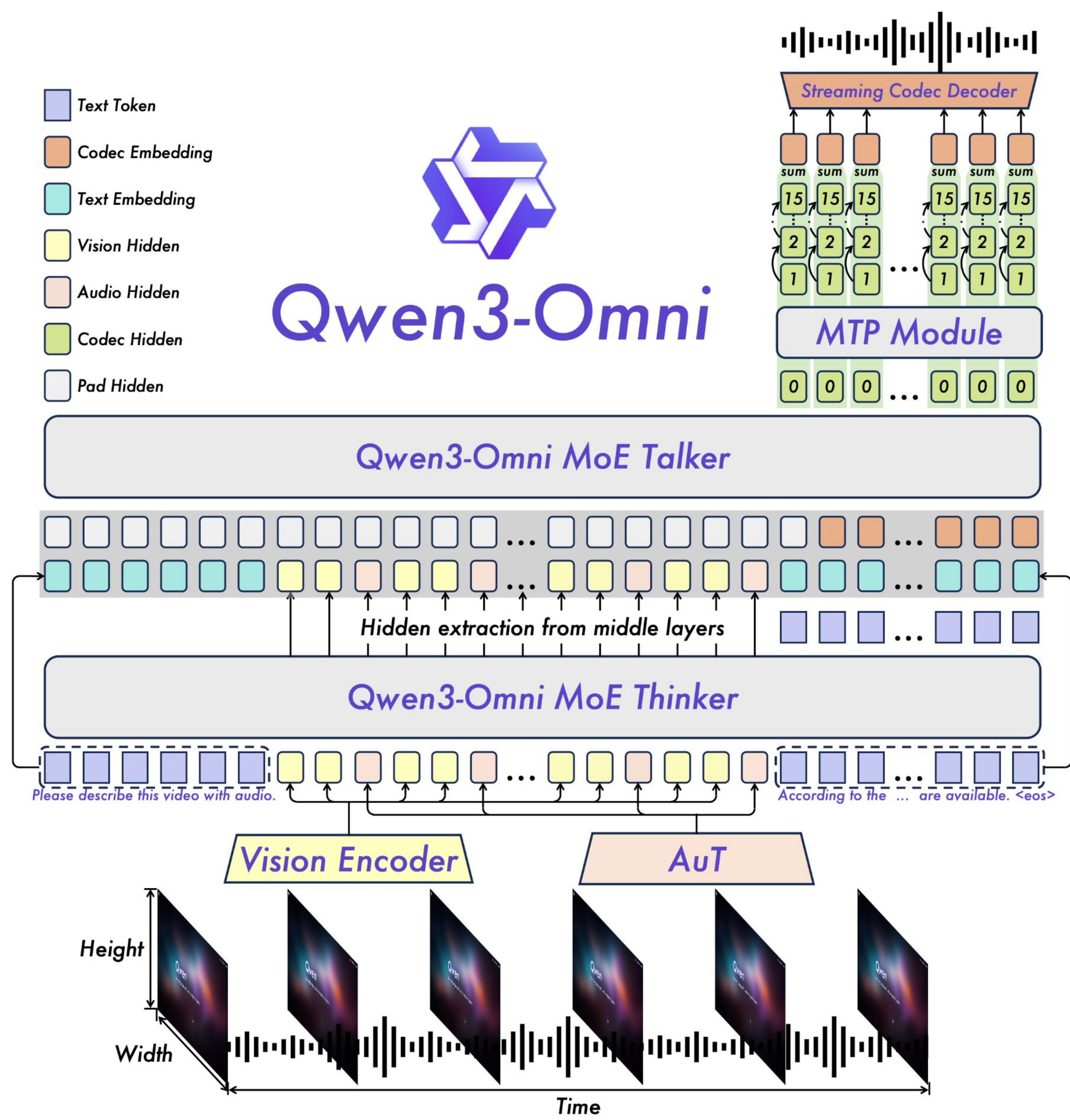
Evolution of AI Models

Expansion into Multimodality and Omni-Modality



Evolution of AI Models

Expansion into Multimodality and Omni-Modality



Evolution of AI Models

Expansion into Multimodality and Omni-Modality



Evolution of AI Models

Expansion into Multimodality and Omni-Modality





Building RAG Agents with LLMs

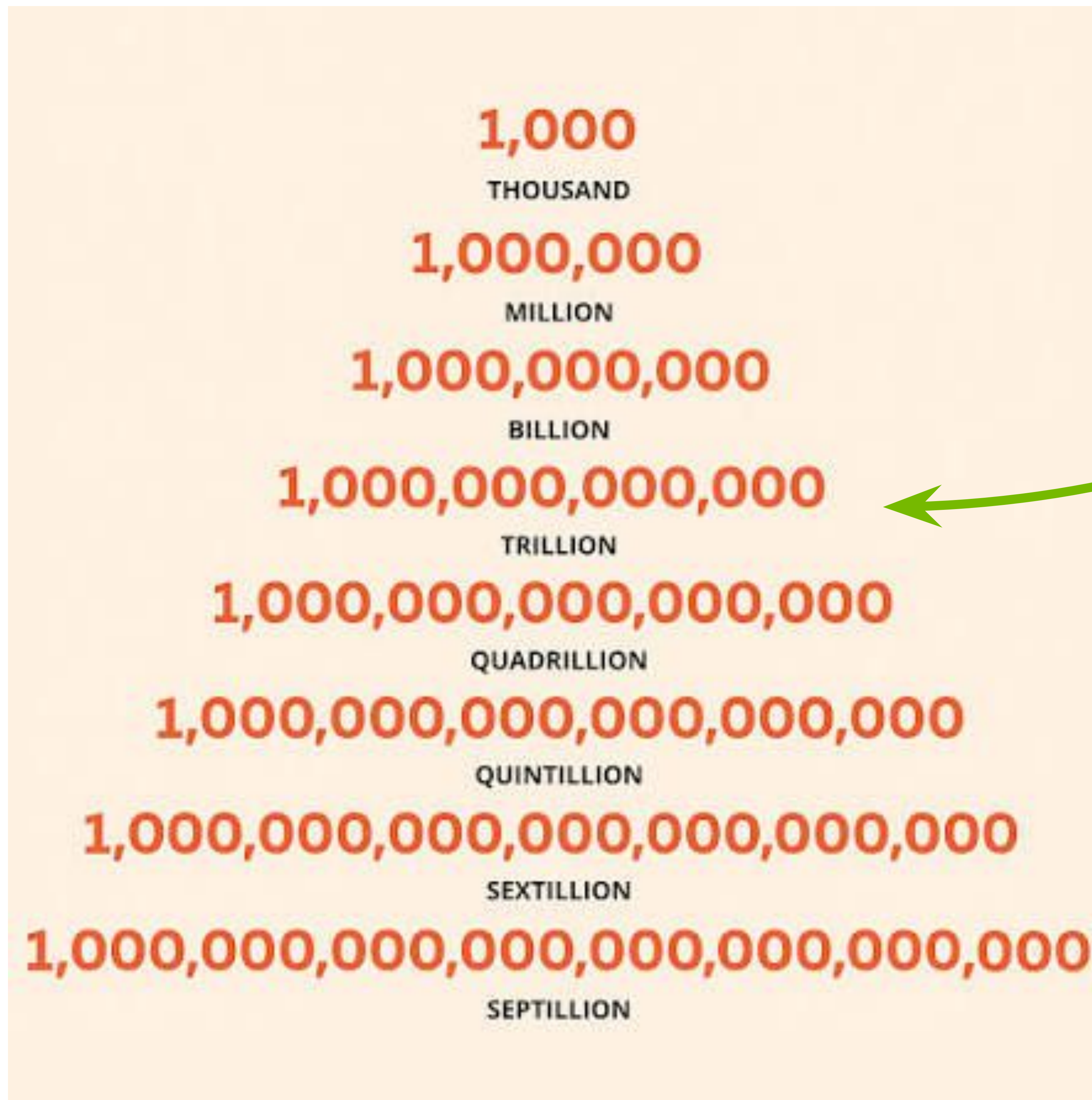
Part 5. GPU Infrastructure

GPU Infrastructure

Recap - Massive Model Parameter



Recap - Massive Model Parameter



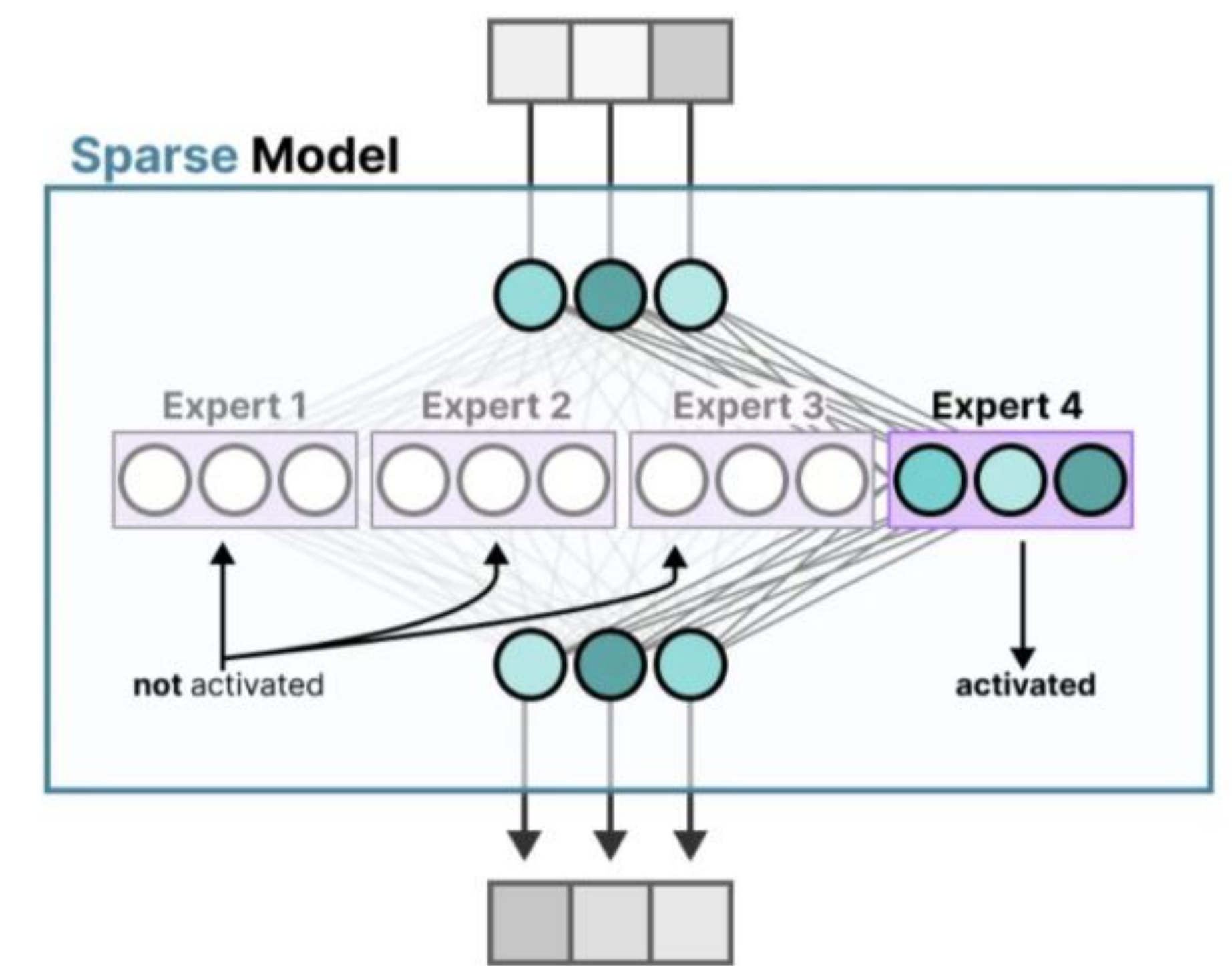
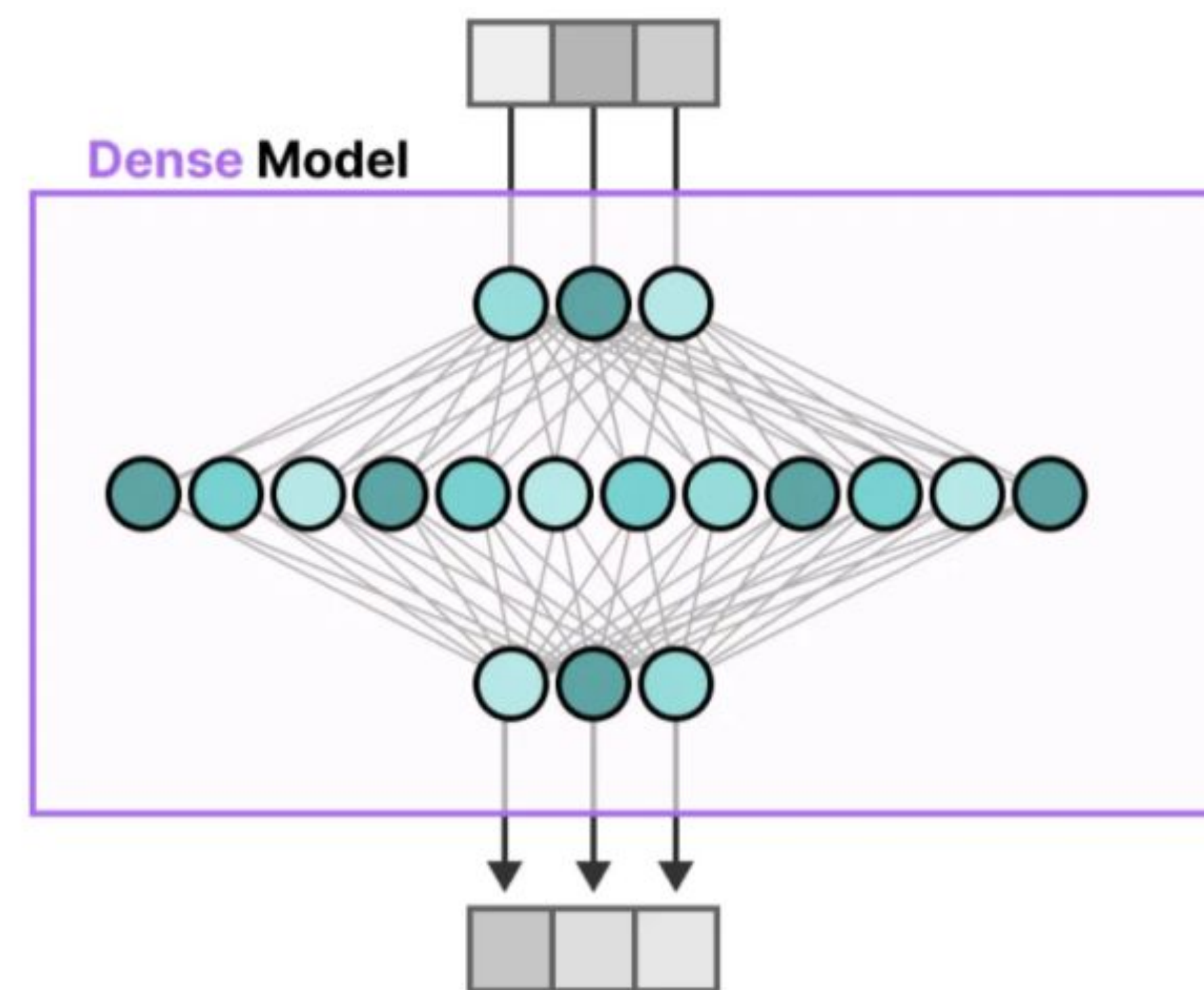
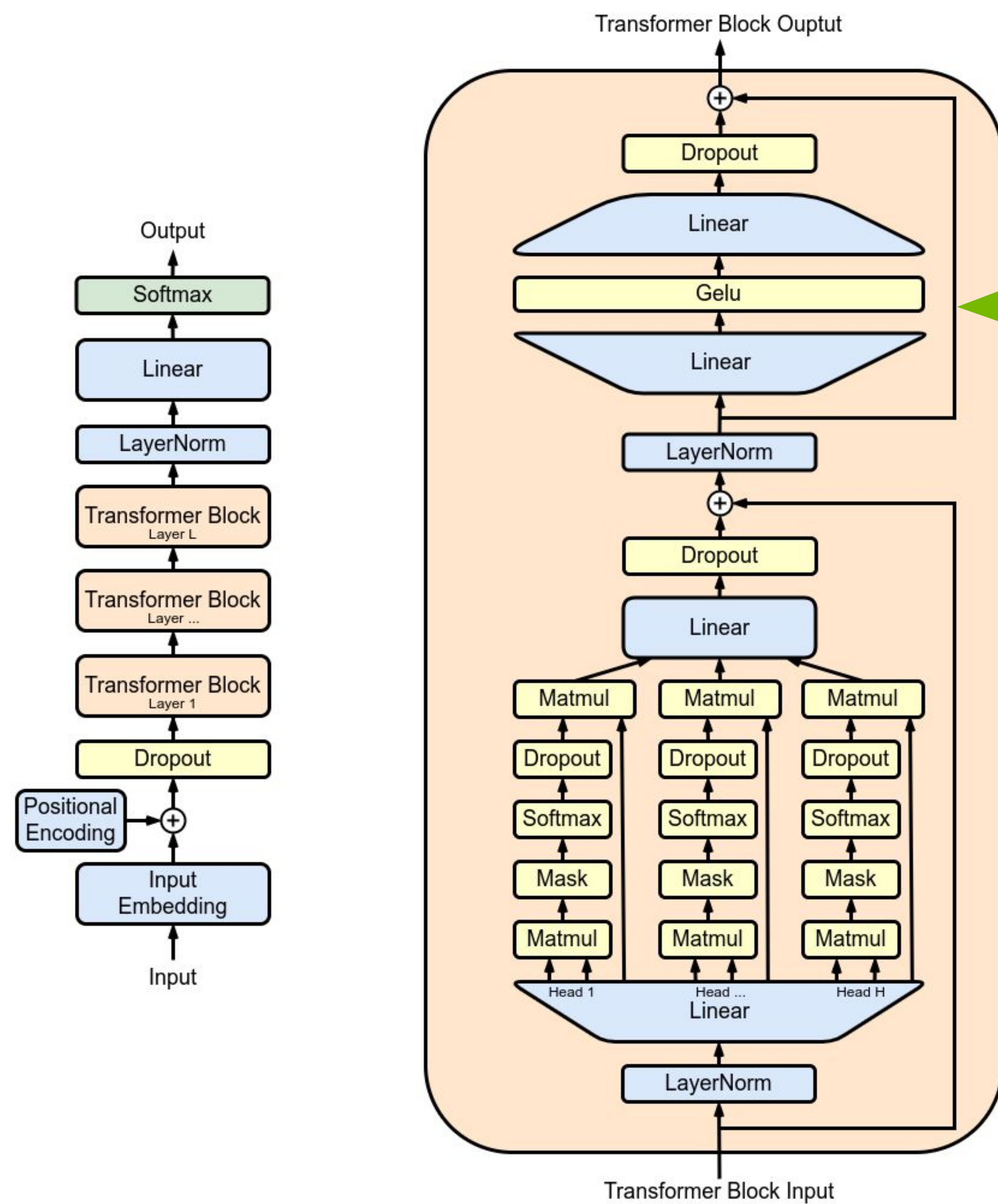
We Are in the Trillion-Parameter Era

How Much Compute Is Required for Training and Inference?

GPU Infrastructure

Modern Architecture - MoE(Mixture of Experts)

Activate Only a Subset of MLP Parameters



GPU Infrastructure

Modern Architecture - MoE(Mixture of Experts)

Gemma 4

updated 13 days ago

Our most intelligent open models to date

 [google/gemma-4-31B-it](#)
 Image-Text-to-Text •  31B • Updated Jul 21 •  8.28M •  •  3.69k

 [google/gemma-4-31B](#)
 Image-Text-to-Text •  33B • Updated Jul 16 •  717k •  517

 [google/gemma-4-26B-A4B-it](#)
 Image-Text-to-Text •  26B • Updated Jul 21 •  8.19M •  •  1.47k

 [google/gemma-4-26B-A4B](#) → **26B Total Parameters, 4B Active Parameters**
 Image-Text-to-Text •  27B • Updated Jul 16 •  80.5k •  392

How to Calculate the Computational Cost of a Model

nvidia/**NVIDIA-Nemotron-3-Nano-30B-A3B-FP8**

Text Generation Transformers Safetensors PyTorch 16 datasets 6 languages nemotron_h nvidia conversational custom_code Eval Results arxiv:2512.20848 arxiv:2512.20856

License: nvidia-nemotron-open-model-license

Model card Files and versions xet Community 29

Deploy Copy to bucket NEW Use this model

NVIDIA-Nemotron-3-Nano-30B-A3B-FP8

Chat Nemotron 3 Nano Paper Read Now! Pre-Training Datasets Available Here

Post-Training Datasets Available Here

Nemotron Developer Page Learn More Here! Discord NVIDIA AI Developer

License NVIDIA Open Model License

Model Overview

Model Developer: NVIDIA Corporation

Model Dates:

September 2025 - December 2025

Data Freshness:

- The post-training data has a cutoff date of November 28, 2025.
- The pre-training data has a cutoff date of June 25, 2025.

Description

Nemotron-Nano-3-30B-A3B-FP8 is a quantized version of Nemotron-Nano-3-30B-A3B and is a large language model (LLM) trained from scratch by NVIDIA, and designed as a unified model for both reasoning and non-reasoning tasks. It responds to user queries and tasks by first generating a reasoning trace and then concluding with a final response. The model's reasoning capabilities can be configured through a flag in the chat template. If the user prefers the model to provide its final

Downloads last month
641,166

Safetensors ⓘ
Model size 32B params Tensor type F32 · BF16 · F8_E4M3 Chat template
Files info

Inference Providers NEW
Text Generation
This model isn't deployed by any Inference Provider. Ask for provider support

Model tree for nvidia/NVIDIA-Nemotron-3-Nano-30B-A3B-FP8 ⓘ
Base model nvidia/NVIDIA-Nemotron-3-Nano-30B-A3B-BF16
Quantized (68) this model

Datasets used to train nvidia/NVIDIA-Nemotron-3-Nano-30B-A3B-FP8

- nvidia/Nemotron-CC-v2
Viewer · Updated Jul 7 · 8.79B · 46.9k · 137
- nvidia/Nemotron-CC-Math-v1
Viewer · Updated Dec 23, 2025 · 190M · 30.3k · 99
- nvidia/Nemotron-Pretraining-Specialized-v1
Viewer · Updated Dec 22, 2025 · 60.7M · 8.15k · 87

Model Parameter Size : B (Billion)

What is the Tensor Type?

GPU Infrastructure

Tensor Type : Bytes per Parameter

Float 32-bit (FP32)

0 1 0 0 0 0 0 0 0 0 1 0 0 1 0 0 1 0 0 0 0 1 1 1 1 1 1 0 1 1 0 1 1

$$(-1)^0 \times 2^1 \times 1.5707964 = 3.1415927410125732$$

higher precision

Float 16-bit (FP16)

0 1 0 0 0 0 1 0 0 1 0 0 1 0 0 0

$$(-1)^0 \times 2^1 \times 1.5703125 = 3.140625$$

lower precision

original value
3.1415927

GPU Infrastructure

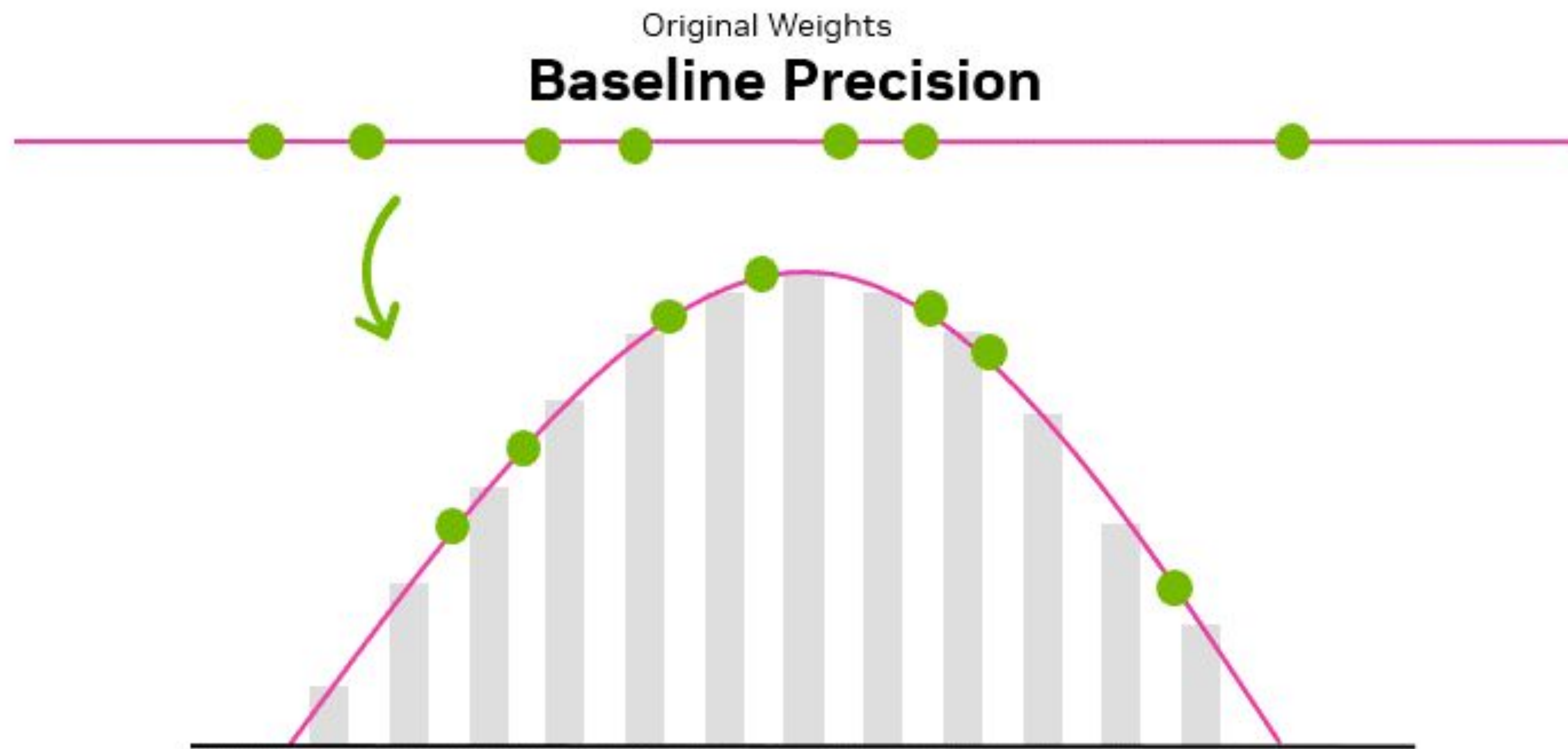
Tensor Type : Bytes per Parameter

Data Type	Total Bits	Representable Range
FP32	32	$\pm 3.4 \times 10^{38}$
FP16	16	$\pm 65,504$
BF16	16	$\pm 3.4 \times 10^{38}$
FP8	8	± 448
FP4	4	-6 to +6
INT8	8	-128 to +127
INT4	4	-8 to +7

Table 1. Summary of data types with bit width, representable ranges, and format descriptions (floating point vs. integer)

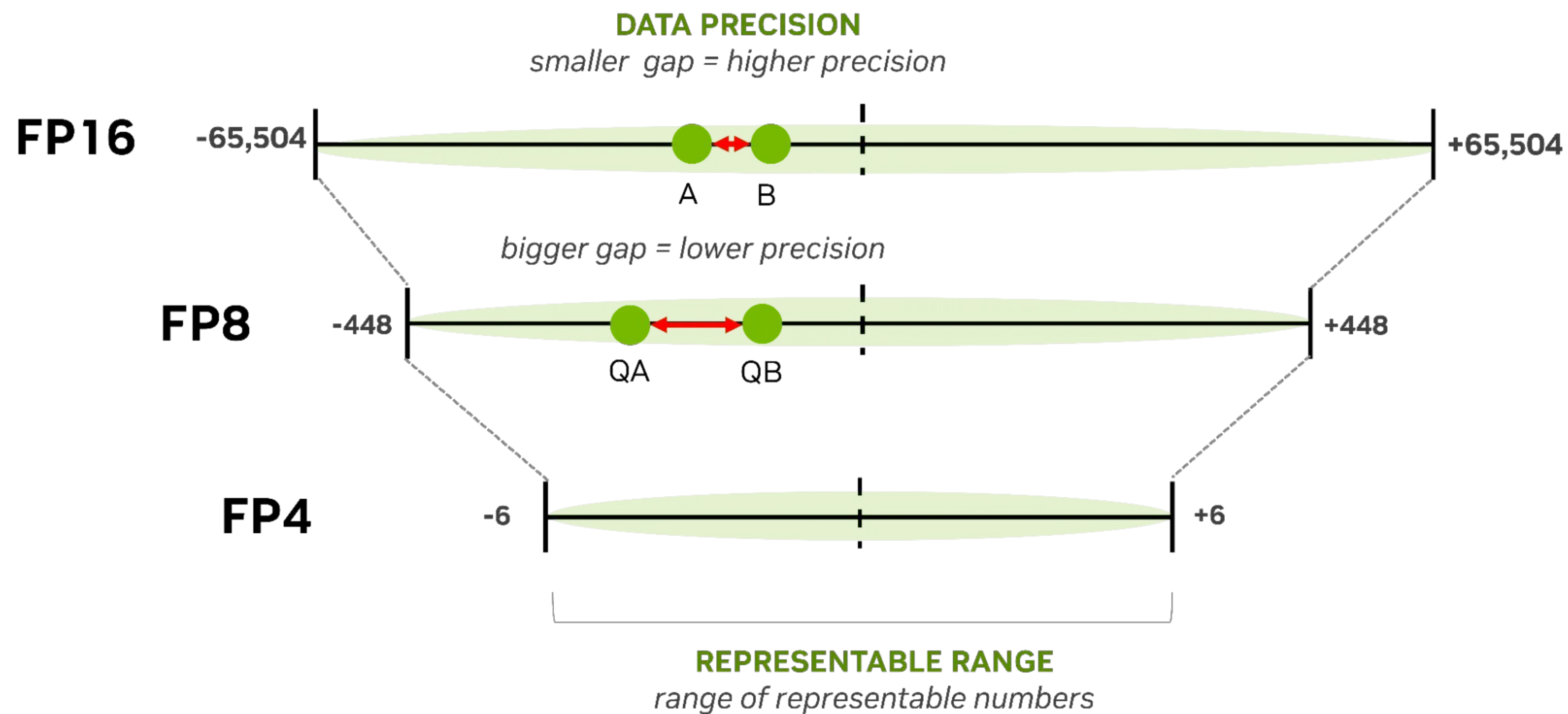
GPU Infrastructure

Tensor Type : Bytes per Parameter



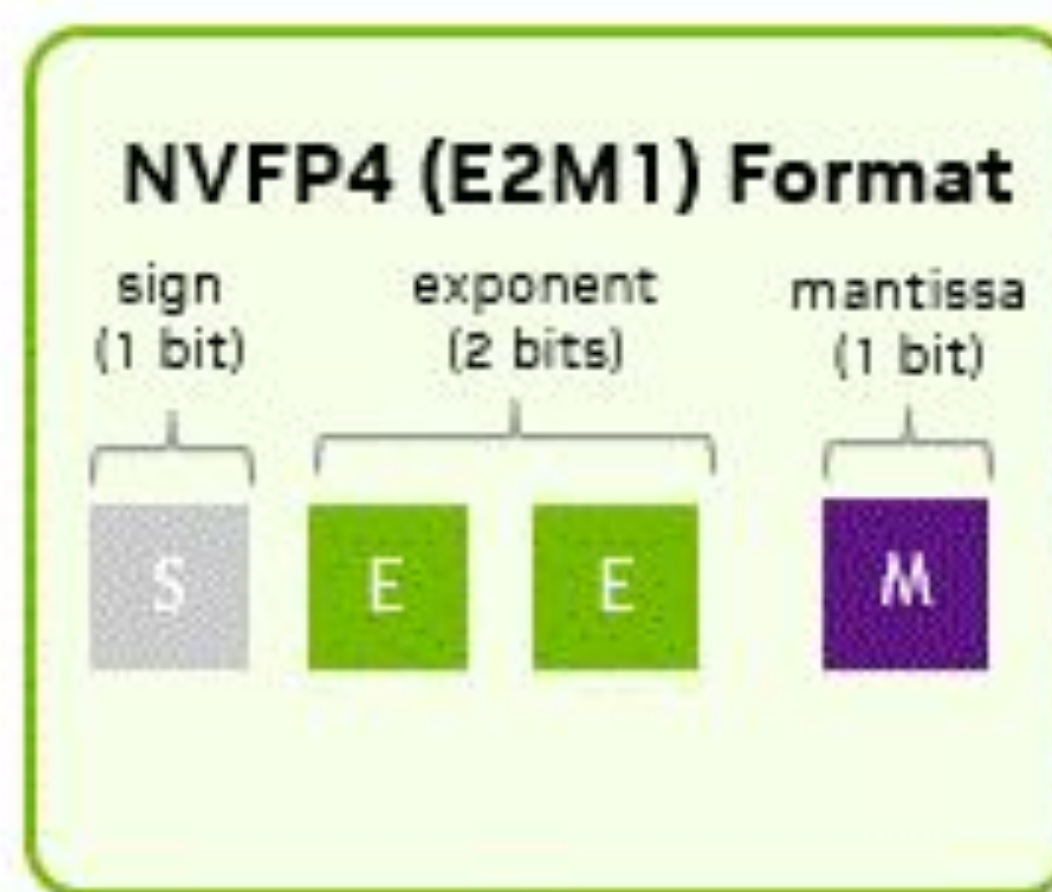
GPU Infrastructure

Tensor Type : Bytes per Parameter



GPU Infrastructure

Tensor Type : Bytes per Parameter



GPU Infrastructure

How to Calculate the Computational Cost of a Model

of Parameter * Tensor Type Size = Total Model Size

ex) 3B Model * int8 = Total 3GB



**2.8T Model * fp32 = Total 11.2TB
(104B Activated → 416GB)**



Can AI models run on a CPU?

→ Yes, but they're slow. Very, very, VERY slow.

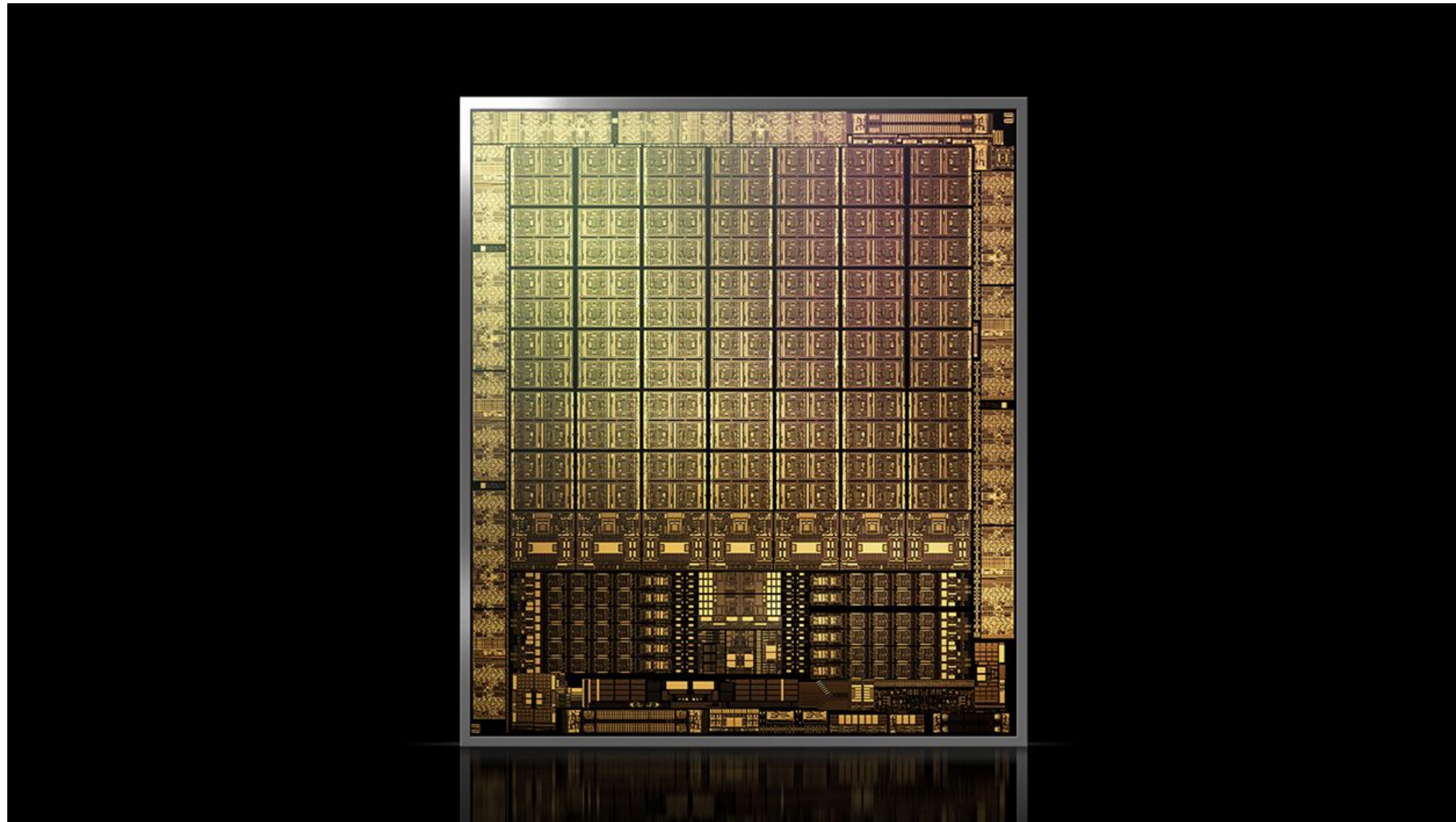
GPU Infrastructure

How to Calculate the Computational Cost of a Model



GPU Infrastructure

How to Calculate the Computational Cost of a Model



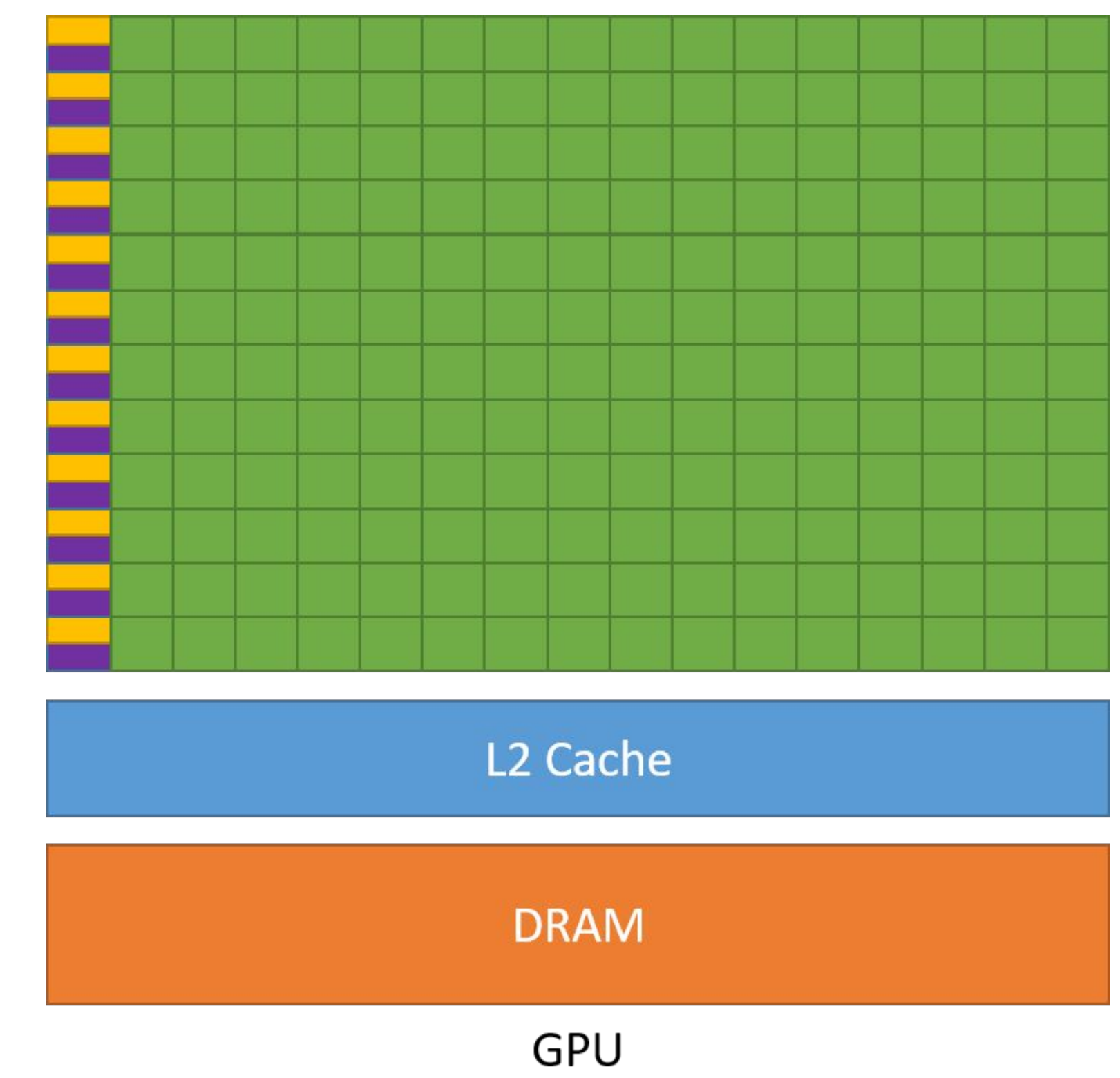
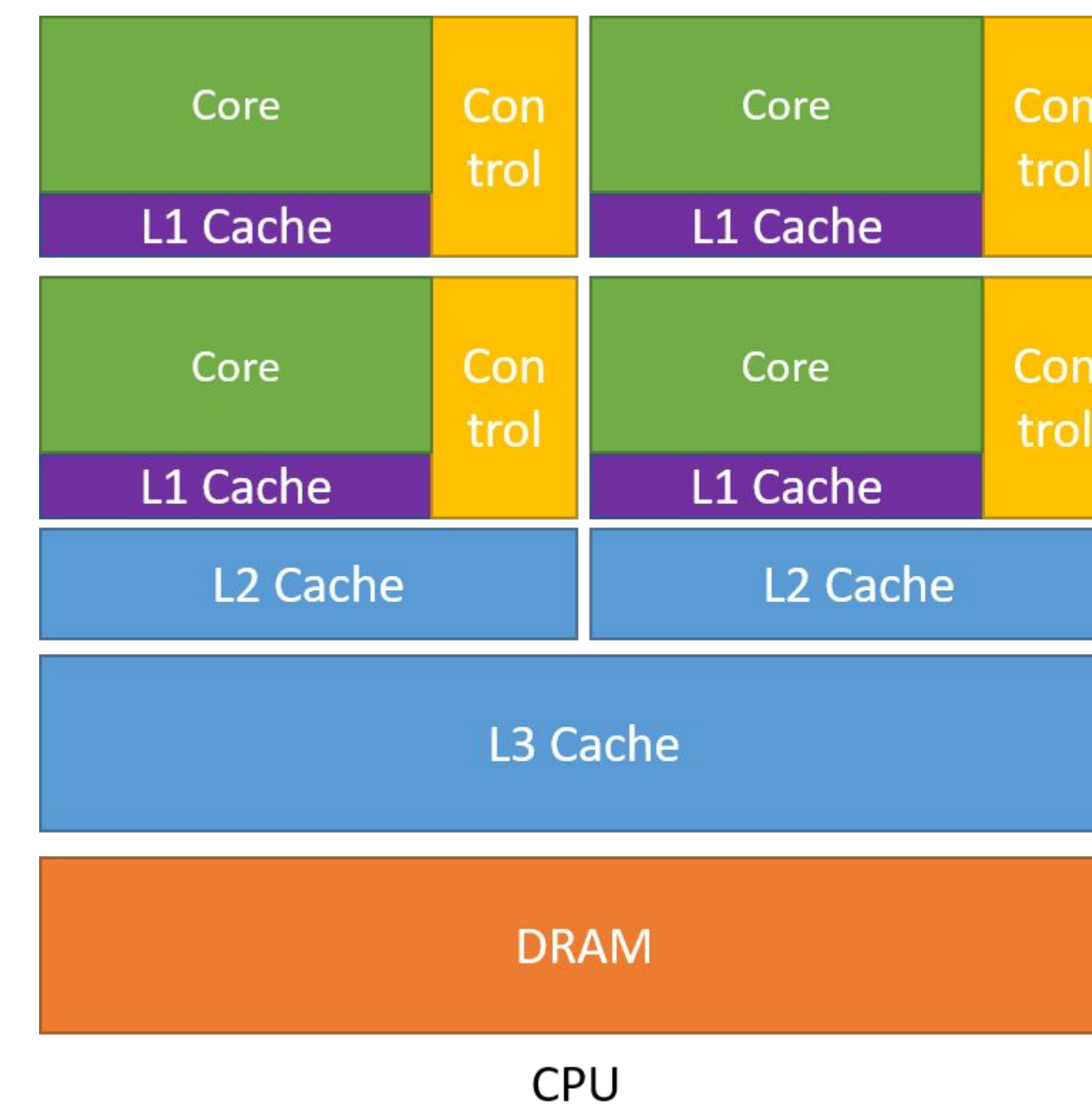
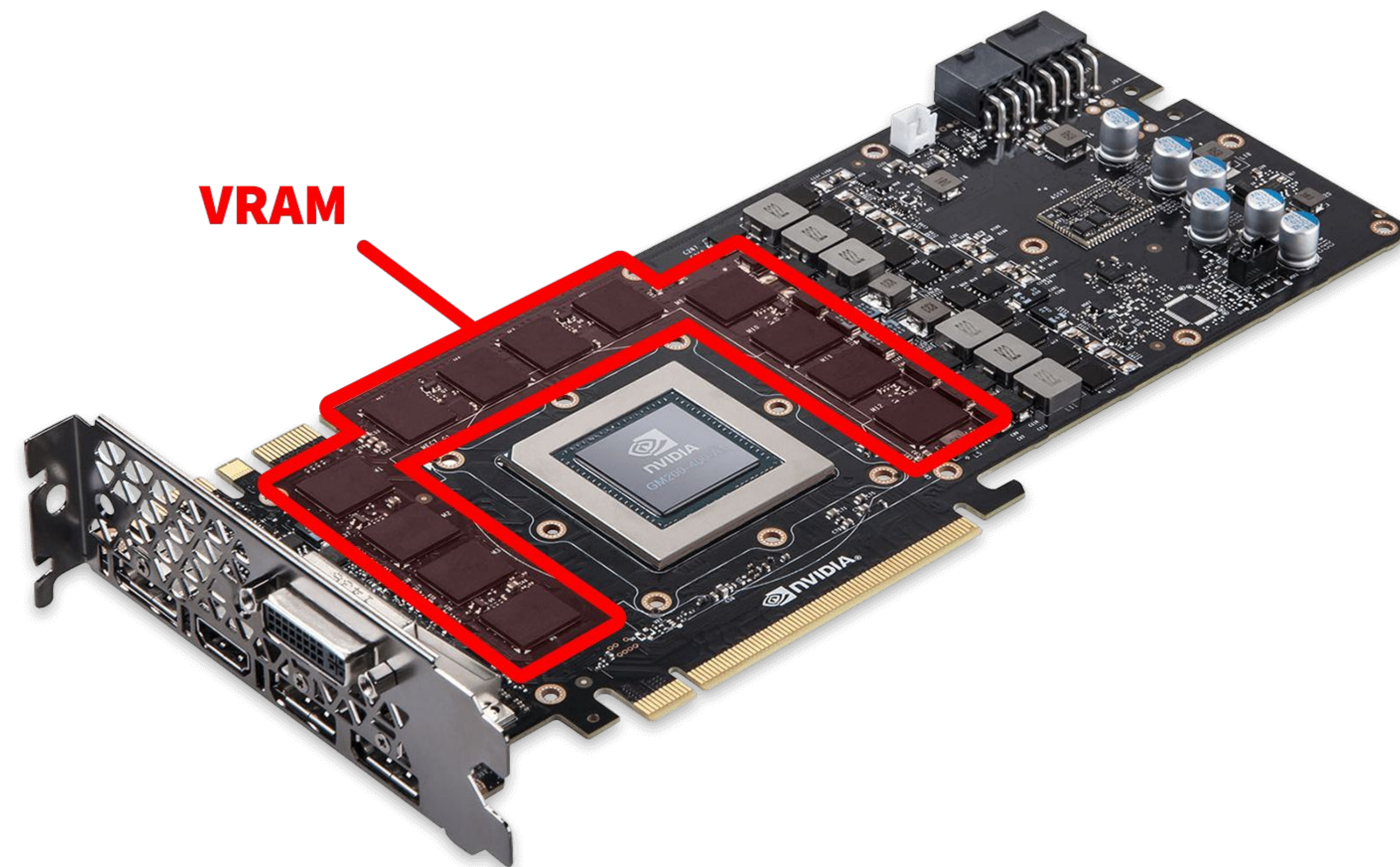
A Powerful Processing Unit — Is It Sufficient?



Memory → GPU
Data Transfer
⇒ New Bottleneck!

GPU Infrastructure

How to Calculate the Computational Cost of a Model

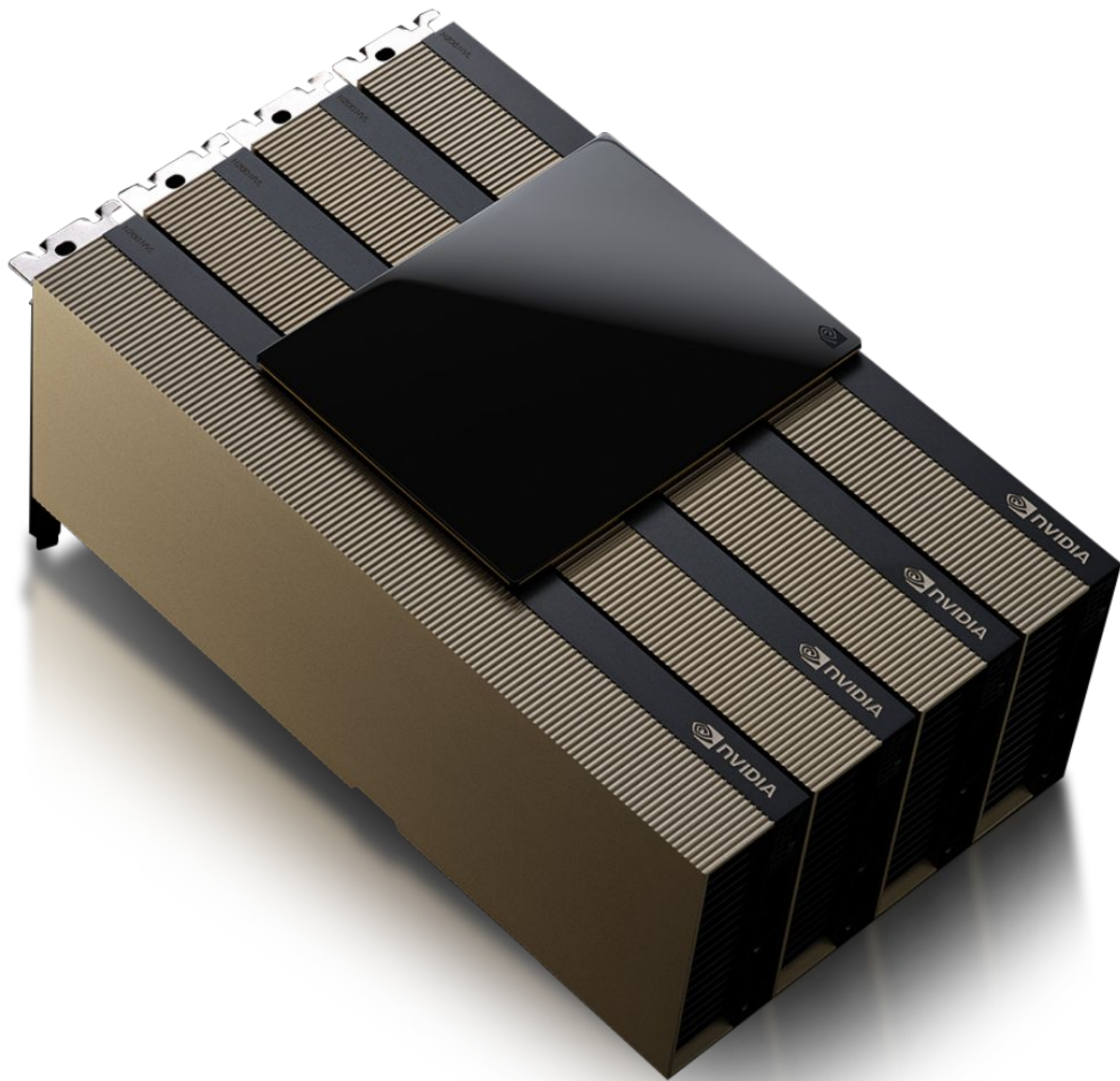


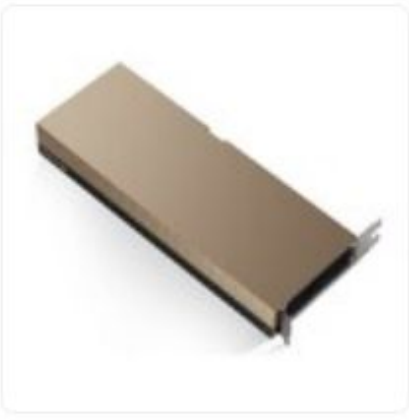
Solution — Store model parameters in GPU memory

GPU Infrastructure

How to Calculate the Computational Cost of a Model

VRAM Matters (On-board GPU Memory)



	10 NVIDIA H100 HBM2e 80GB PCIe H100 / PCIe x16 / 최대 AI TOPS: 3026(sparsity) / HBM(2세대) ECC / VRAM 대역폭: 2000 GB/s / 지원정보: 멀티VGA 지원 / 사용전력: 350W 24.04. 등록 의견 2 관심 대량구매	45,314,600원 +
	7 NVIDIA H100 HBM3 94GB NVL H100 / PCIe5.0×16 / 스트림 프로세서: 16896 / 최대 AI TOPS: 3958(x2, sparsity) / HBM3 / VRAM 대역폭: 3900 GB/s / 지원정보: 멀티VGA 지원 / 사용전력: 400W 24.05. 등록 의견 20 관심 대량구매	44,738,950원 +
	14 NVIDIA A100 HBM2e 80GB PCIe A100 / PCIe4.0 / 스트림 프로세서: 6912 / 최대 AI TOPS: 1248(sparsity) / HBM2 / VRAM 대역폭: 2000 GB/s / 지원정보: 멀티VGA 지원 / 사용전력: 300W / FP16 Tensor Core: 312 TFLOPs / FP32: 19.5 TFLOPs 23.07. 등록 관심 대량구매	40,459,760원 +
	12 NVIDIA RTX PRO 6000 Blackwell 서버 에디션 D7 96GB (병행수입) RTX PRO 6000 Blackwell / PCIe5.0×16 / 가로(길이): 266.7mm / 스트림 프로세서: 24064 / GDDR7 ECC / 출력단자: DP2.1 / 지원정보: 8K 지원 / 사용전력: 600W / 두께: 39mm 25.07. 등록 관심 대량구매	27,499,990원 +
	5 NVIDIA RTX PRO 6000 Blackwell 서버 에디션 D7 96GB 서린 (벌크) RTX PRO 6000 Blackwell / PCIe5.0×16 / 가로(길이): 266.7mm / 스트림 프로세서: 24064 / GDDR7 ECC / 출력단자: DP2.1 / 지원정보: 8K 지원 / 사용전력: 600W / 무팬 / 두께: 39mm 25.07. 등록 브랜드로그 관심 대량구매	27,200,000원 +



Building RAG Agents with LLMs

THANK YOU SO MUCH!

Jae Y. CHOI

NVIDIA DLI Certified Instructor & University Ambassador

본 자료의 저작권은 Jae Y. CHOI (jaeyeong2022@gmail.com)에게 있습니다.

본 자료는 자유롭게 이용할 수 있으나, 사용 시 저작자 및 출처를 반드시 표기해 주시기 바랍니다.

Copyright © Jae Y. CHOI (jaeyeong2022@gmail.com).

This material may be freely used, provided that the author and source are properly credited.